[Lecture Notebook] Data Visualization

MGS3701 Data Mining, Spring 2025

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Sun, 25 May 2025

Table 3.2

```
housing.df <- read.csv("data/BostonHousing.csv")
head(housing.df, 9)
```

```
      CRIM   ZN INDUS CHAS   NOX    RM   AGE    DIS RAD TAX PTRATIO LSTAT MEDV
1 0.00632 18.0  2.31    0 0.538 6.575  65.2 4.0900   1 296    15.3  4.98 24.0
2 0.02731  0.0  7.07    0 0.469 6.421  78.9 4.9671   2 242    17.8  9.14 21.6
3 0.02729  0.0  7.07    0 0.469 7.185  61.1 4.9671   2 242    17.8  4.03 34.7
4 0.03237  0.0  2.18    0 0.458 6.998  45.8 6.0622   3 222    18.7  2.94 33.4
5 0.06905  0.0  2.18    0 0.458 7.147  54.2 6.0622   3 222    18.7  5.33 36.2
6 0.02985  0.0  2.18    0 0.458 6.430  58.7 6.0622   3 222    18.7  5.21 28.7
7 0.08829 12.5  7.87    0 0.524 6.012  66.6 5.5605   5 311    15.2 12.43 22.9
8 0.14455 12.5  7.87    0 0.524 6.172  96.1 5.9505   5 311    15.2 19.15 27.1
9 0.21124 12.5  7.87    0 0.524 5.631 100.0 6.0821   5 311    15.2 29.93 16.5
  CAT..MEDV
1         0
2         0
3         1
4         1
5         1
6         0
7         0
8         0
9         0
```

Figure 3.1

Amtrack ridership by year

```
Amtrak.df <- read.csv("data/Amtrak data.csv")

# use time series analysis
library(forecast)
```

```
Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo
```

```
Registered S3 methods overwritten by 'forecast':
  method  from
  head.ts stats
  tail.ts stats
```

```
ridership.ts <- ts(Amtrak.df$Ridership, start = c(1991, 1), end = c(2004, 3), freq = 12)
plot(ridership.ts, xlab = "Year", ylab = "Ridership (in 000s)", ylim = c(1300, 2300))
```

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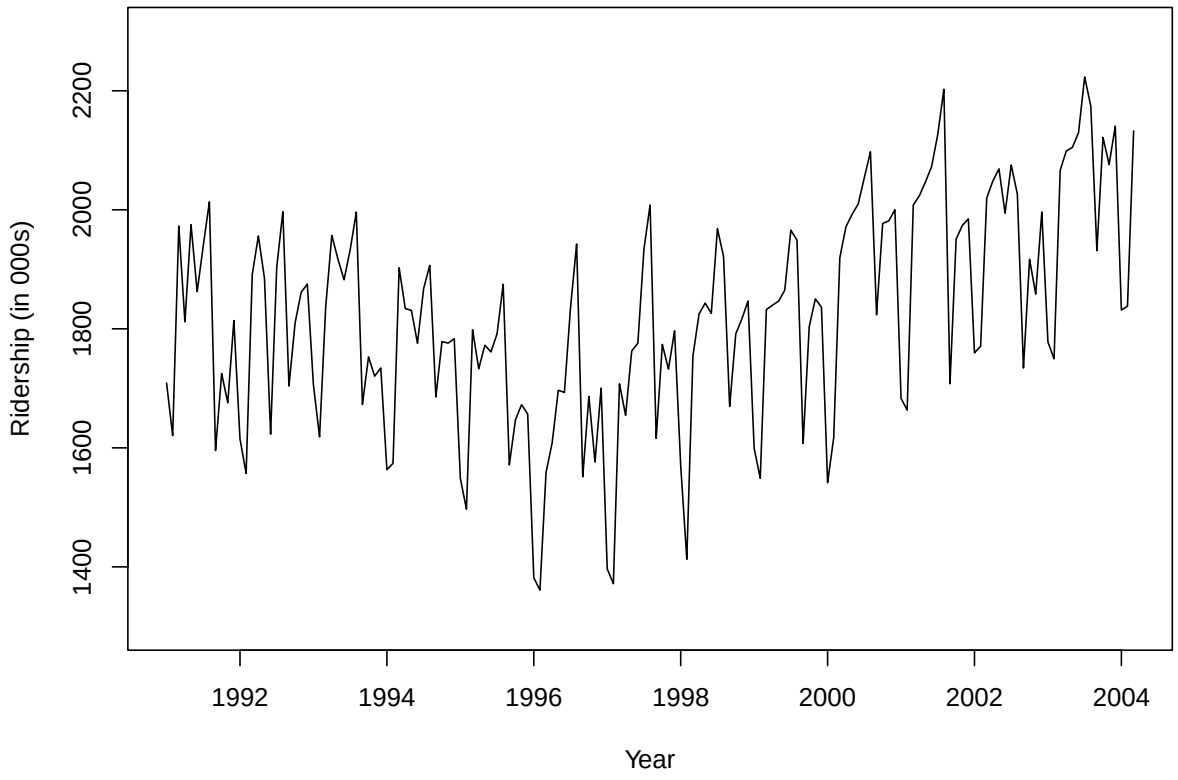

Boston housing data

```
housing.df <- read.csv("data/BostonHousing.csv")

## scatter plot with axes names
plot(housing.df$MEDV ~ housing.df$LSTAT, xlab = "MDEV", ylab = "LSTAT")
```

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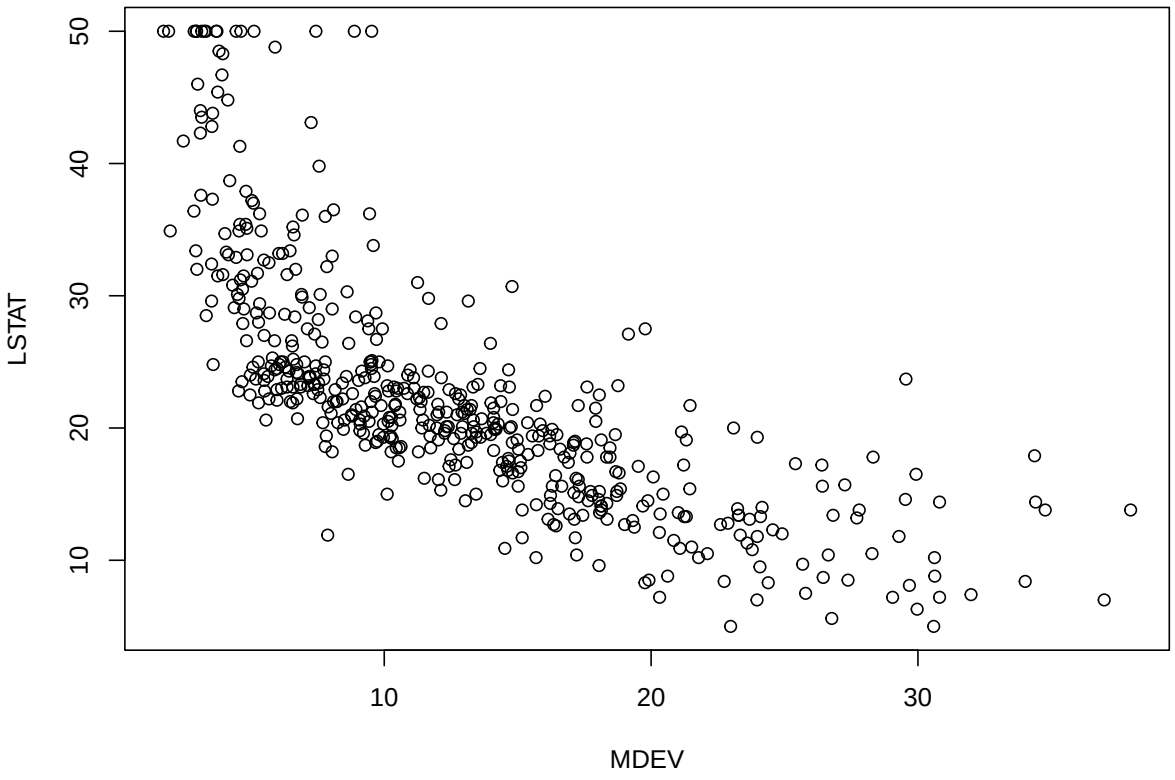

37

```
# alternative plot with ggplot
library(ggplot2)
ggplot(housing.df) + geom_point(aes(x = LSTAT, y = MEDV), colour = "navy", alpha = 0.7)
```

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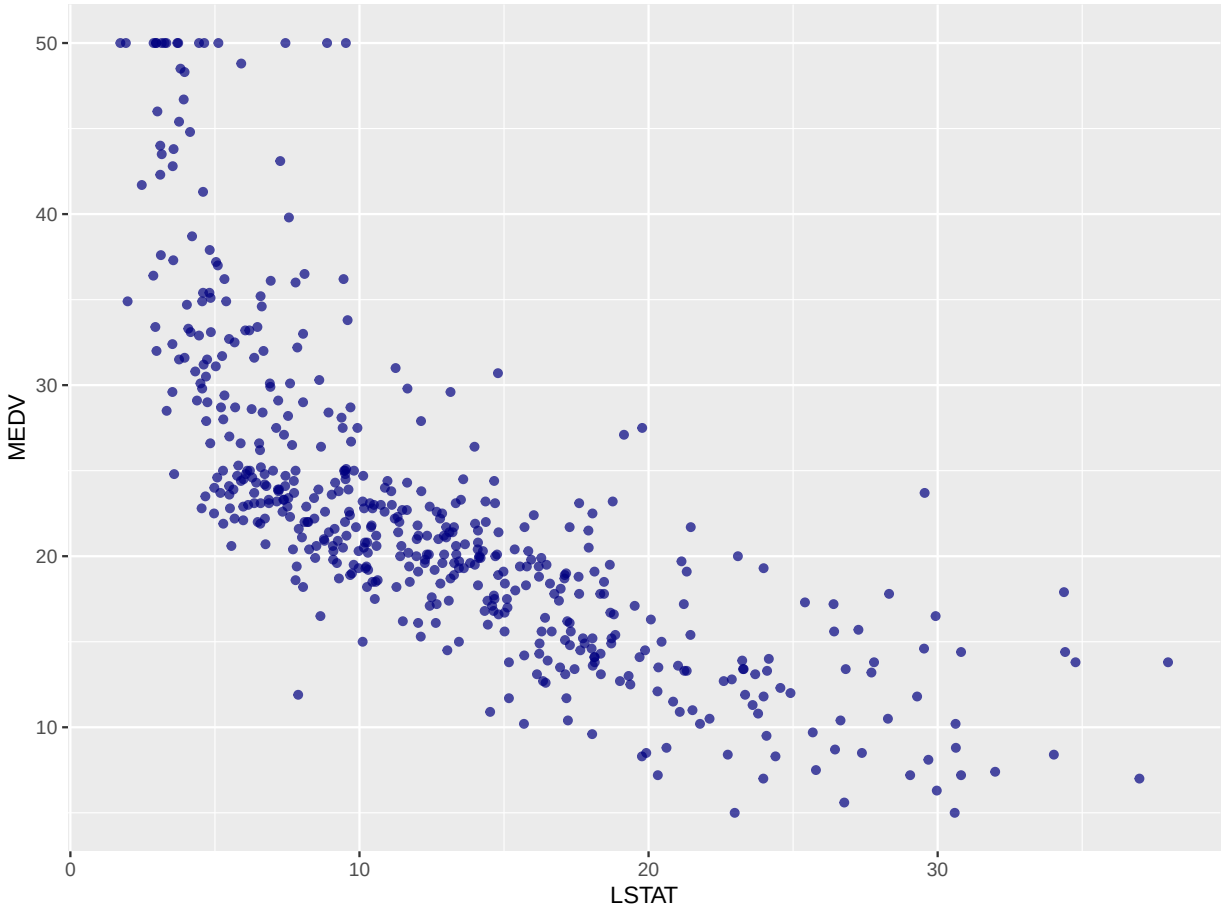

barchart of CHAS vs. mean MEDV

```
# compute mean MEDV per CHAS = (0, 1)
data.for.plot <- aggregate(housing.df$MEDV, by = list(housing.df$CHAS), FUN = mean)
names(data.for.plot) <- c("CHAS", "MeanMEDV")
barplot(data.for.plot$MeanMEDV,  names.arg = data.for.plot$CHAS,
        xlab = "CHAS", ylab = "Avg. MEDV")
```

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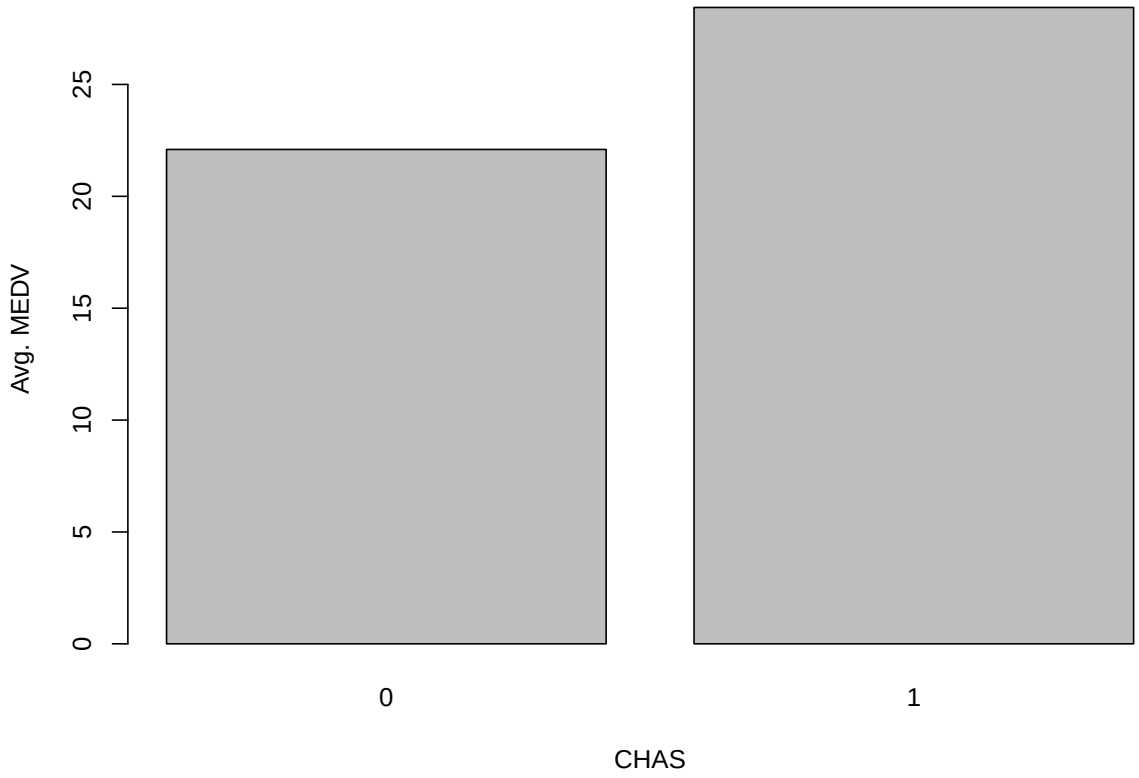

40

```
# alternative plot with ggplot
ggplot(data.for.plot) + geom_bar(aes(x = CHAS, y = MeanMEDV), stat = "identity")
```

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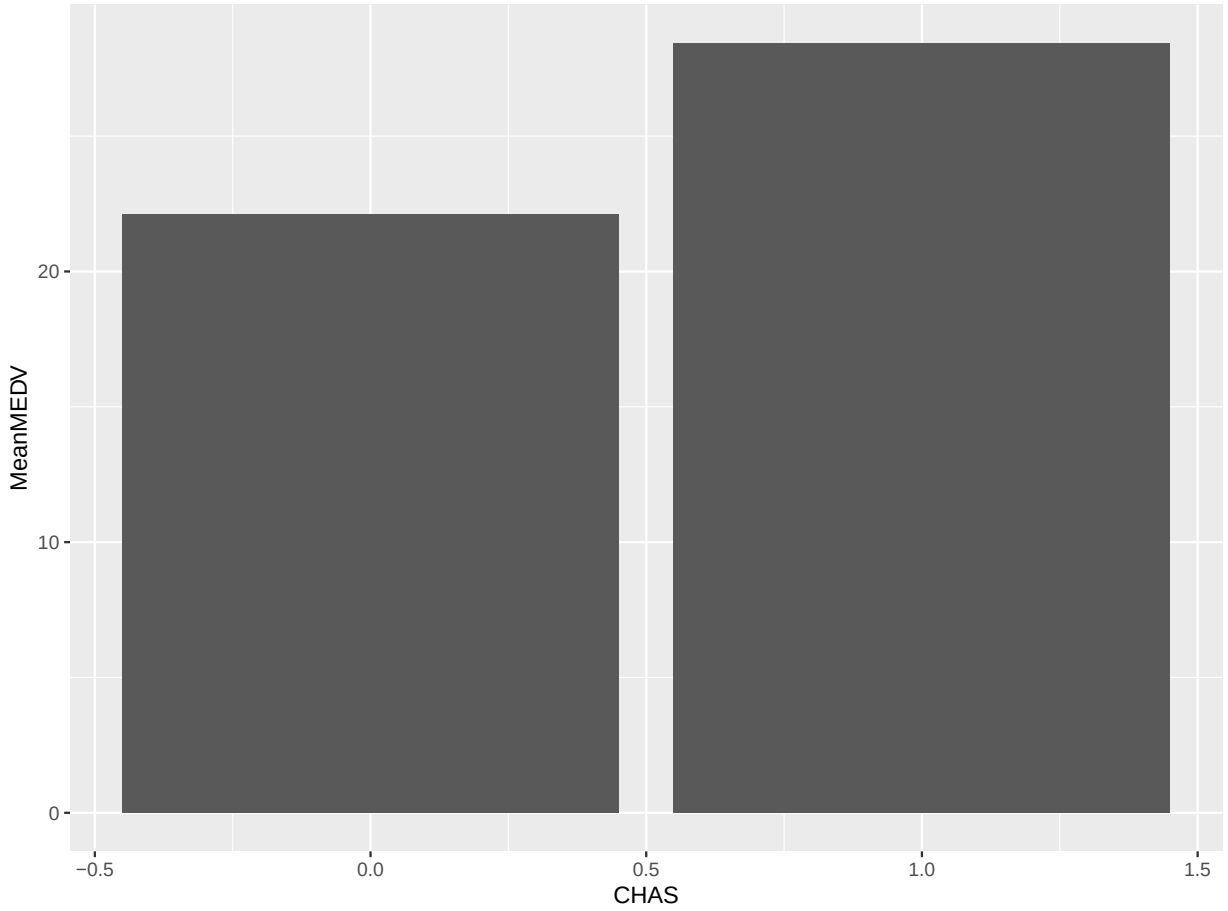

barchart of CHAS vs. % CAT.MEDV

```
data.for.plot <- aggregate(housing.df$CAT..MEDV, by = list(housing.df$CHAS), FUN = mean)
names(data.for.plot) <- c("CHAS", "MeanCATMEDV")
barplot(data.for.plot$MeanCATMEDV * 100,  names.arg = data.for.plot$CHAS,
        xlab = "CHAS", ylab = "% of CAT.MEDV")
```

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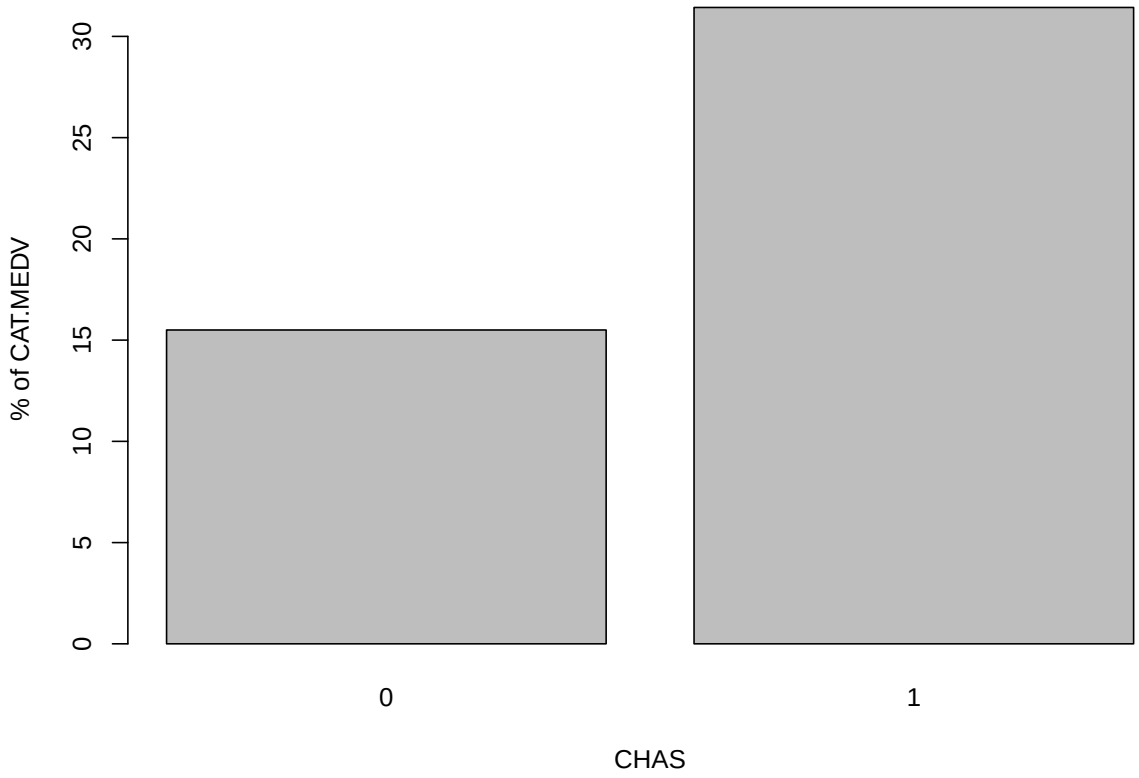

43

44 Figure 3.2

45 histogram of MEDV

```
hist(housing.df$MEDV, xlab = "MEDV")
```

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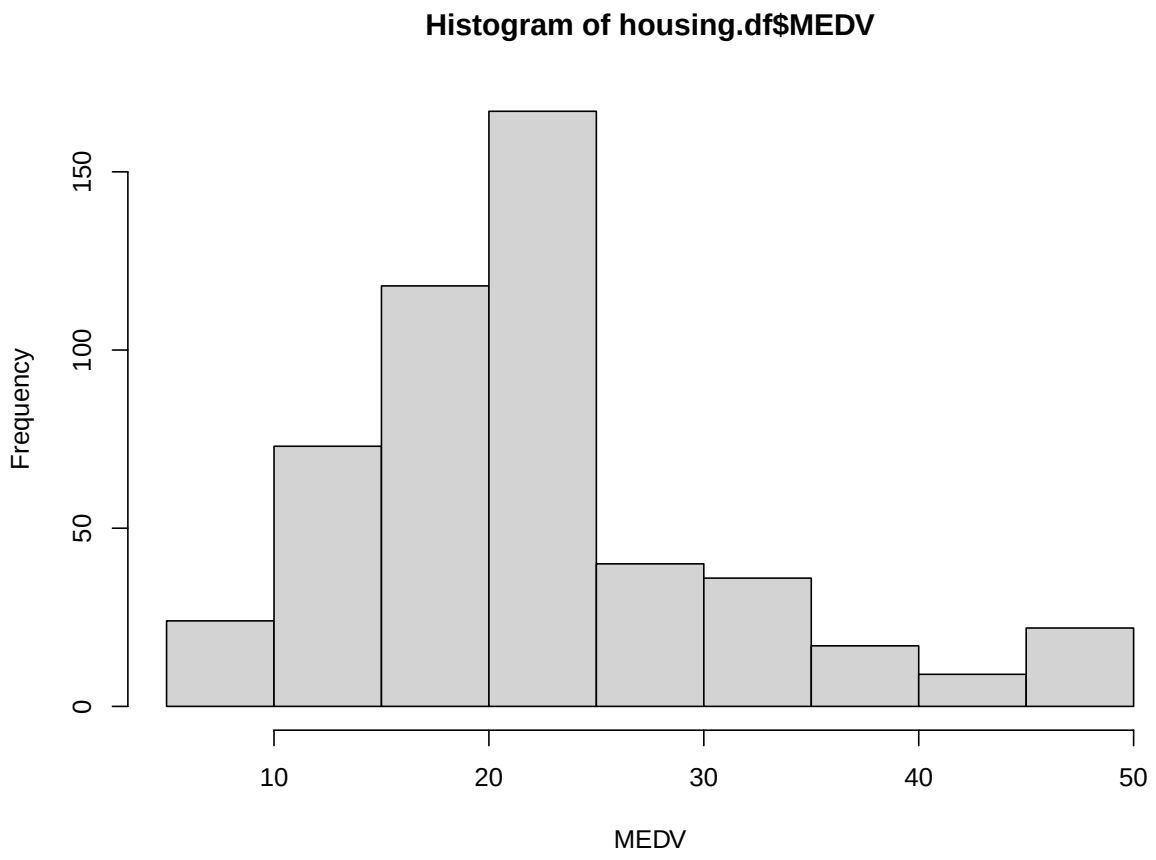

46

47 **alternative plot with ggplot**

```
library(ggplot2)
ggplot(housing.df) + geom_histogram(aes(x = MEDV), binwidth = 5)
```

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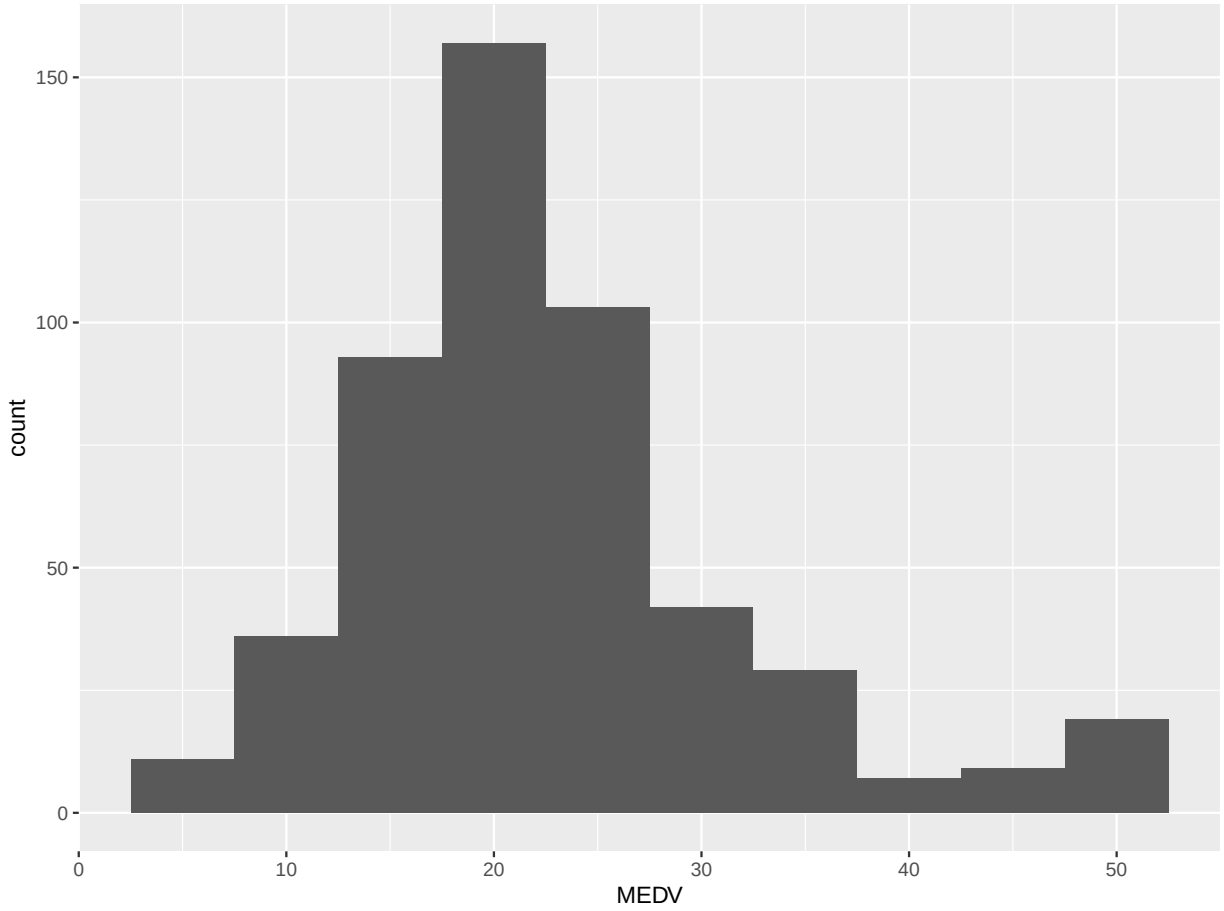

boxplot of MEDV for different values of CHAS

```
boxplot(housing.df$MEDV ~ housing.df$CHAS, xlab = "CHAS", ylab = "MEDV")
```

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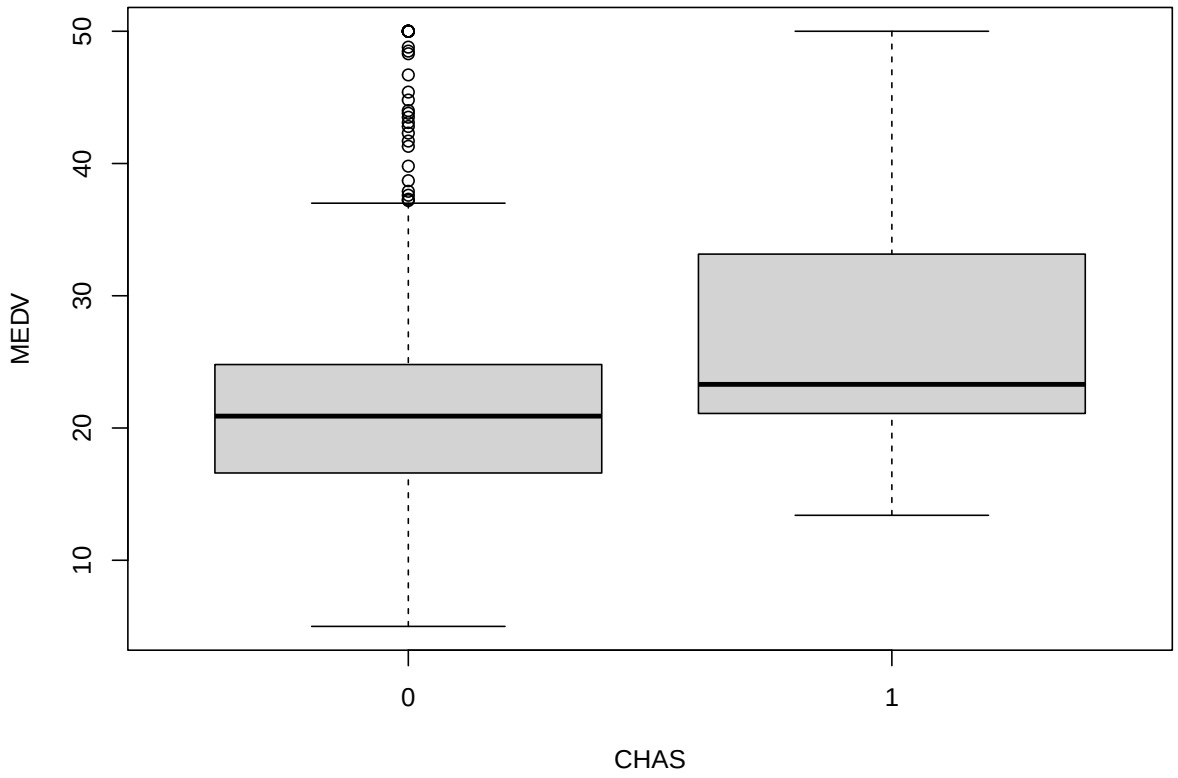

50

```
# alternative plot with ggplot
ggplot(housing.df) + geom_boxplot(aes(x = as.factor(CHAS), y = MEDV)) + xlab("CHAS")
```

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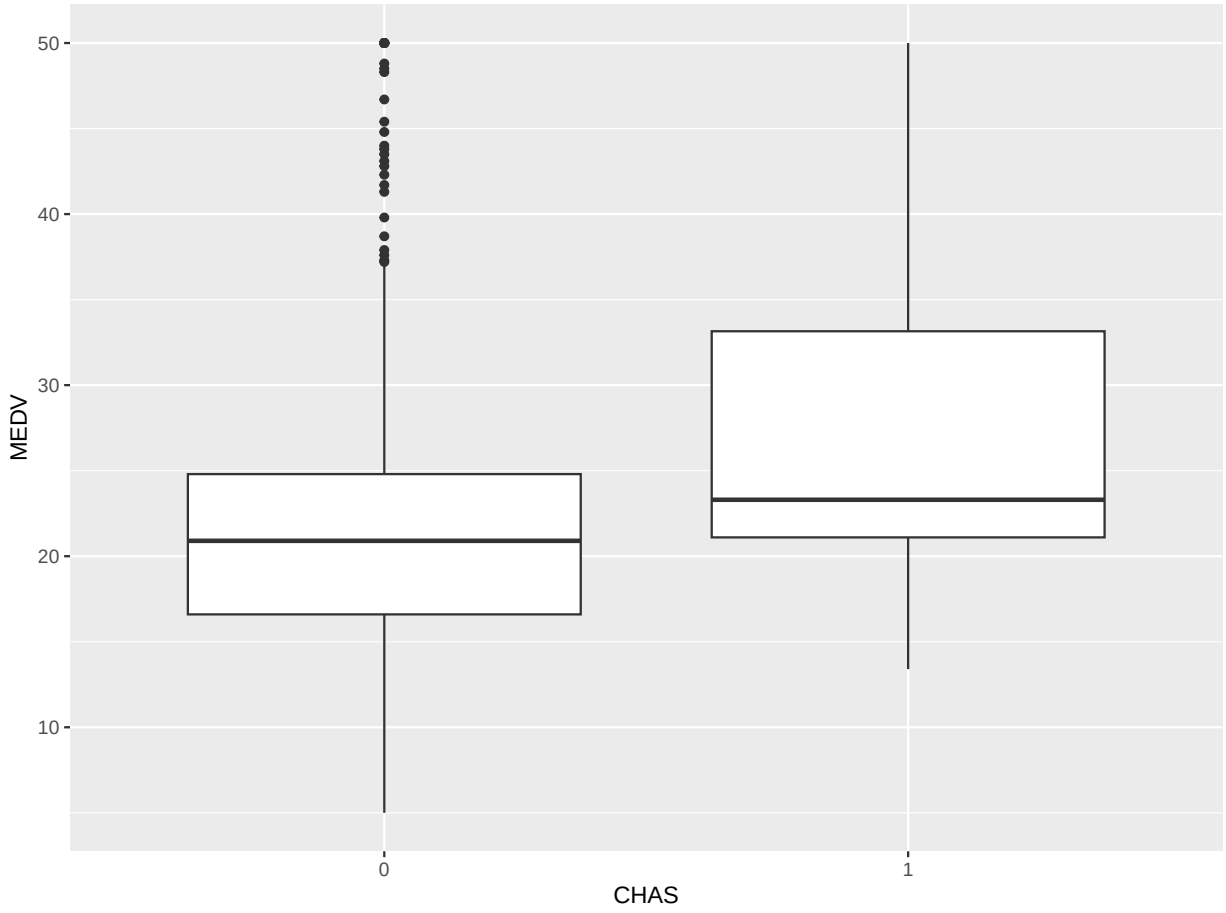

Figure 3.3

side-by-side boxplots

```
# use par() to split the plots into panels.
par(mfcol = c(1, 4))
boxplot(housing.df$NOX ~ housing.df$CAT..MEDV, xlab = "CAT.MEDV", ylab = "NOX")
boxplot(housing.df$LSTAT ~ housing.df$CAT..MEDV, xlab = "CAT.MEDV", ylab = "LSTAT")
boxplot(housing.df$PTRATIO ~ housing.df$CAT..MEDV, xlab = "CAT.MEDV", ylab = "PTRATIO")
boxplot(housing.df$INDUS ~ housing.df$CAT..MEDV, xlab = "CAT.MEDV", ylab = "INDUS")
```

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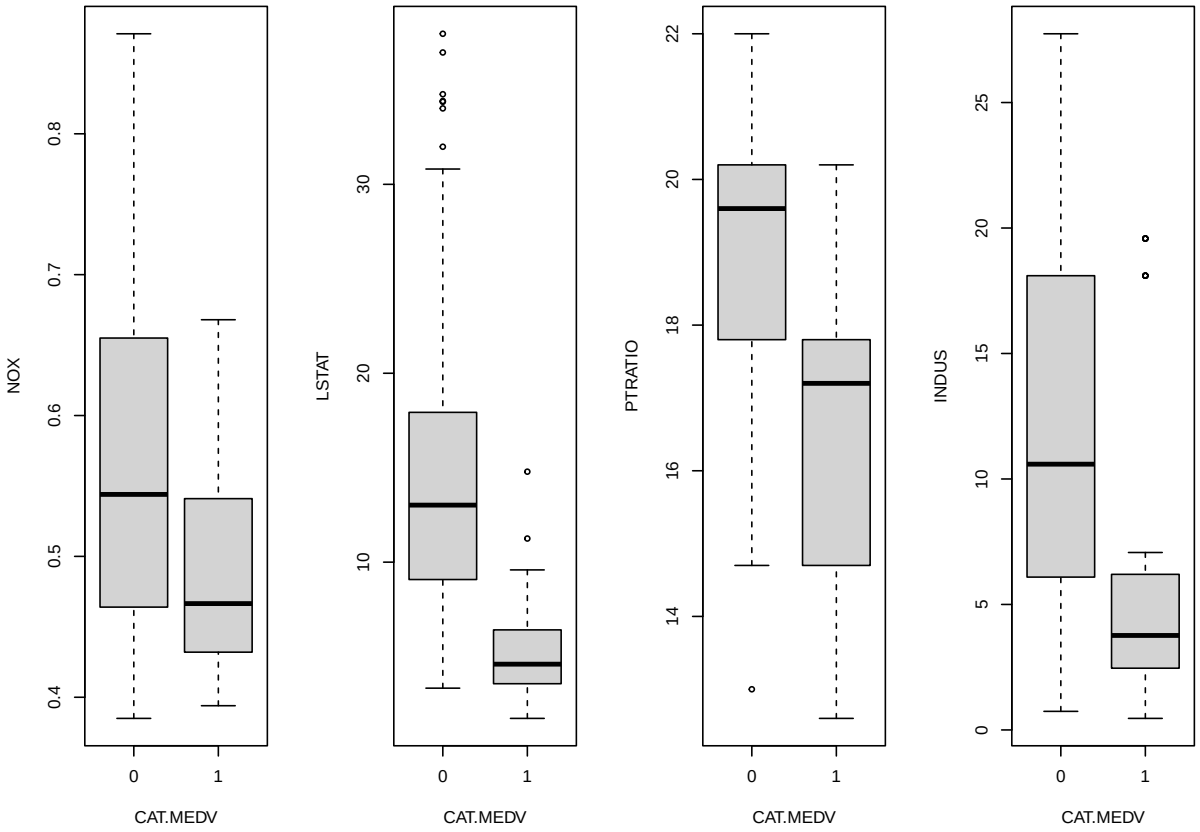

Figure 3.4

simple heatmap of correlations (without values)

```
heatmap(cor(housing.df), Rowv = NA, Colv = NA)
```

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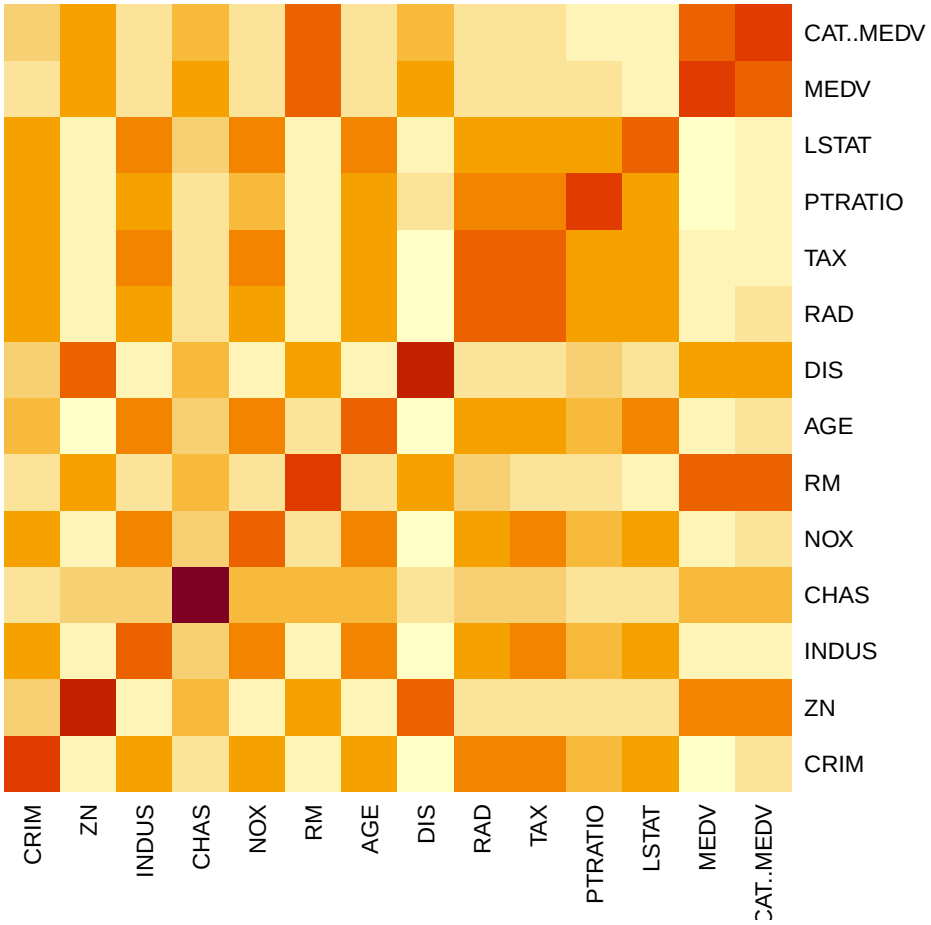

heatmap with values

```
library(gplots)
```

```
Attaching package: 'gplots'

The following object is masked from 'package:stats':

    lowess
```

```
heatmap.2(cor(housing.df), Rowv = FALSE, Colv = FALSE, dendrogram = "none",
          cellnote = round(cor(housing.df),2),
          notecol = "black", key = FALSE, trace = 'none', margins = c(10,10))
```

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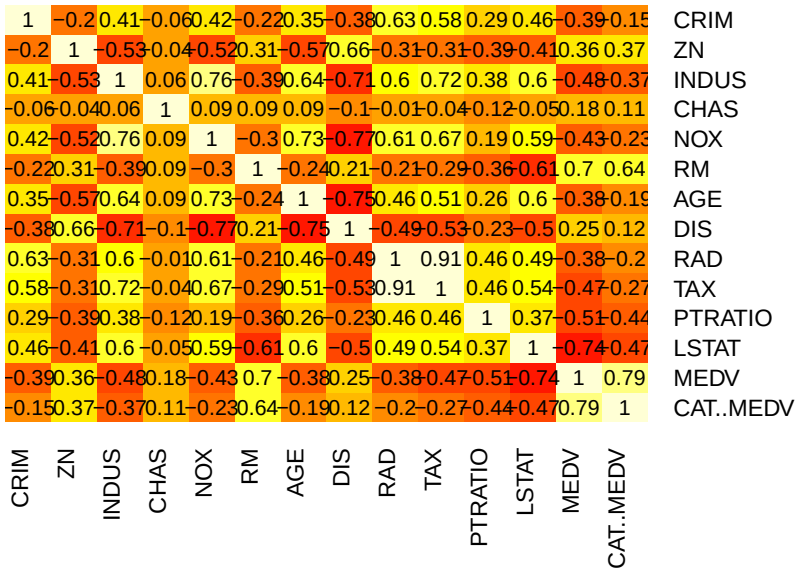

alternative plot with ggplot

```
library(ggplot2)
library(reshape) # to generate input for the plot
```

```
Attaching package: 'reshape'

The following object is masked from 'package:lubridate':

    stamp

The following object is masked from 'package:dplyr':

    rename
```

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```
The following objects are masked from 'package:tidyr':

    expand, smiths
```

```
cor.mat <- round(cor(housing.df),2) # rounded correlation matrix
melted.cor.mat <- melt(cor.mat)
ggplot(melted.cor.mat, aes(x = X1, y = X2, fill = value)) +
  geom_tile() +
  geom_text(aes(x = X1, y = X2, label = value))
```

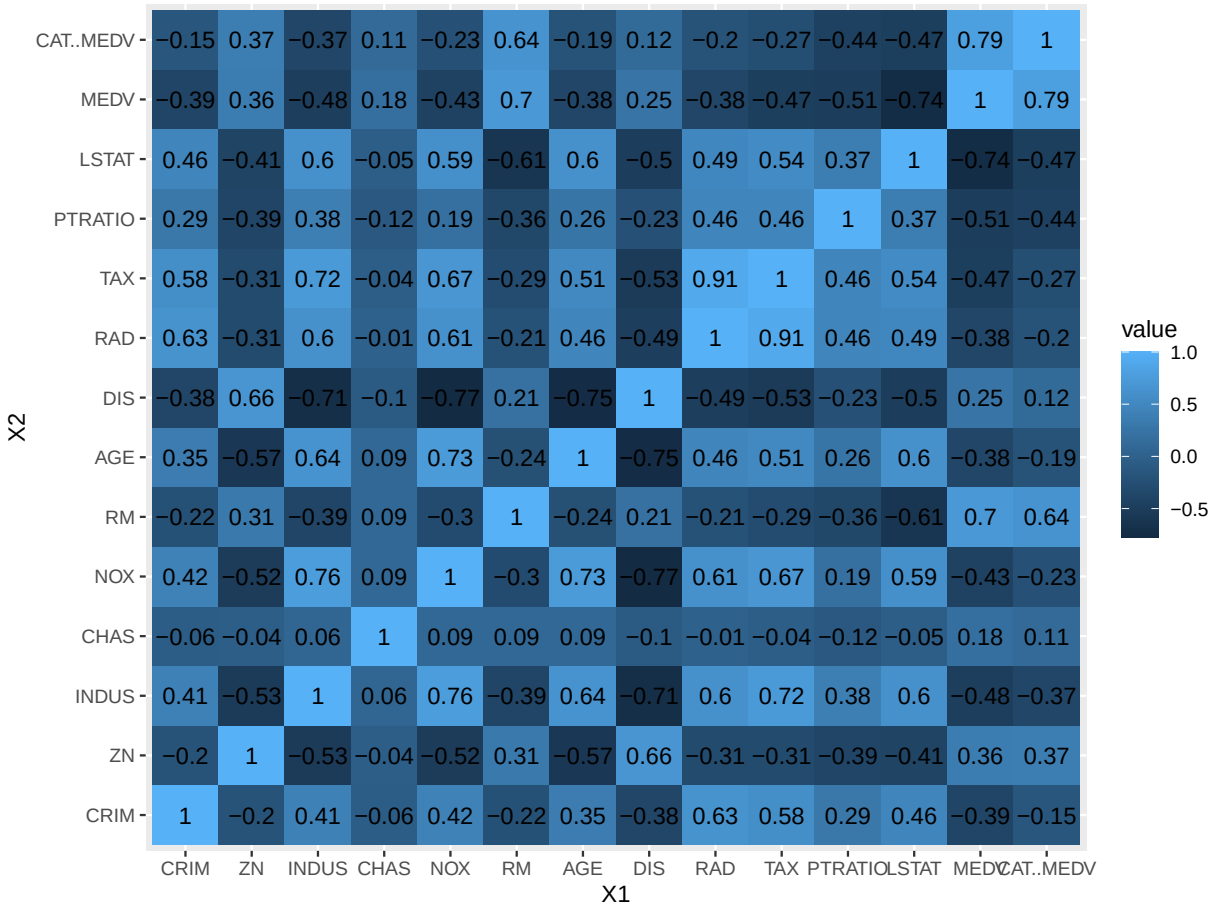

Figure 3.5

```
# replace dataFrame with your data.
# is.na() returns a Boolean (TRUE/FALSE) output indicating the location of missing
# values.
```

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```
# multiplying the Boolean value by 1 converts the output into binary (0/1).
#heatmap(1 * is.na(dataFrame), Rowv = NA, Colv = NA)
```

79 ## Figure 3.6

80 **Housing data scatter plot with color CAT..MEDV**

```
## color plot
par(mfcol = c(1,1), xpd=TRUE) # allow legend to be displayed outside of plot area
plot(housing.df$NOX ~ housing.df$LSTAT, ylab = "NOX", xlab = "LSTAT",
     col = ifelse(housing.df$CAT..MEDV == 1, "black", "gray"))
# add legend outside of plotting area
# In legend() use argument inset =  to control the location of the legend relative
# to the plot.
legend("topleft", inset=c(0, -0.2),
       legend = c("CAT.MEDV = 1", "CAT.MEDV = 0"), col = c("black", "gray"),
       pch = 1, cex = 0.5)
```

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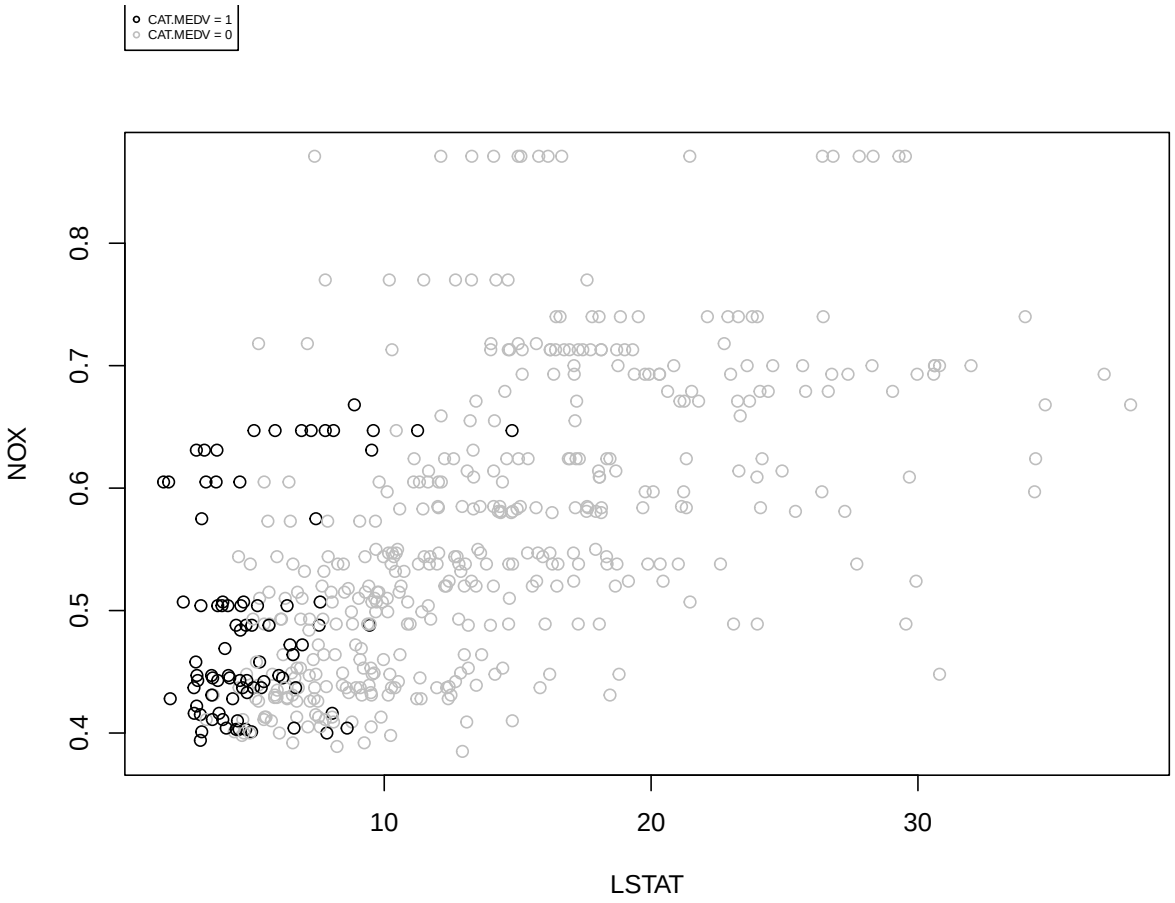

81

```
# alternative plot with ggplot
library(ggplot2)
ggplot(housing.df, aes(y = NOX, x = LSTAT, colour= CAT..MEDV)) +
  geom_point(alpha = 0.6)
```

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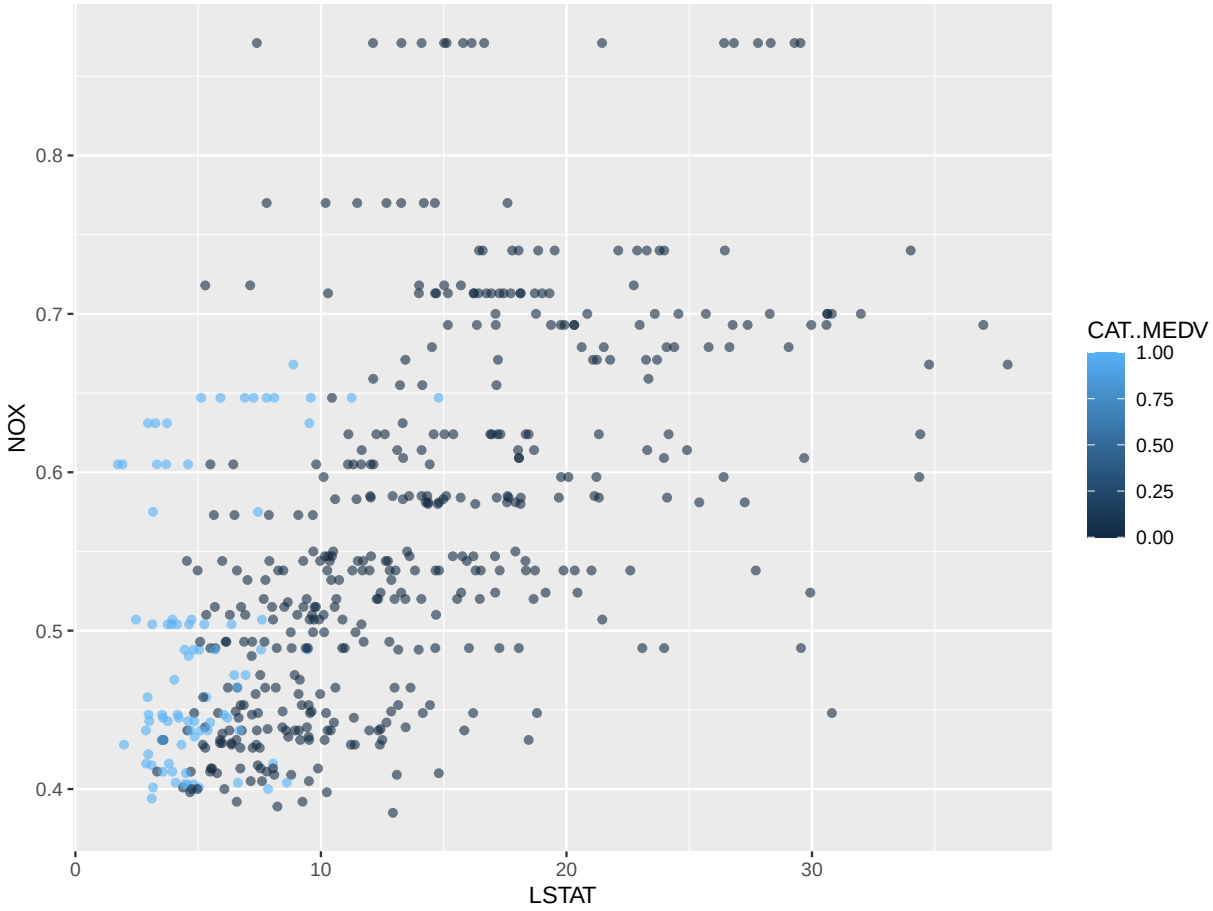

panel plots

```
# compute mean MEDV per RAD and CHAS
# In aggregate() use argument drop = FALSE to include all combinations
# (exiting and missing) of RAD X CHAS.
data.for.plot <- aggregate(housing.df$MEDV, by = list(housing.df$RAD, housing.df$CHAS),
                           FUN = mean, drop = FALSE)
names(data.for.plot) <- c("RAD", "CHAS", "meanMEDV")
# plot the data
par(mfcol = c(2,1))
barplot(height = data.for.plot$meanMEDV[data.for.plot$CHAS == 0],
        names.arg = data.for.plot$RAD[data.for.plot$CHAS == 0],
        xlab = "RAD", ylab = "Avg. MEDV", main = "CHAS = 0")
barplot(height = data.for.plot$meanMEDV[data.for.plot$CHAS == 1],
        names.arg = data.for.plot$RAD[data.for.plot$CHAS == 1],
        xlab = "RAD", ylab = "Avg. MEDV", main = "CHAS = 1")
```

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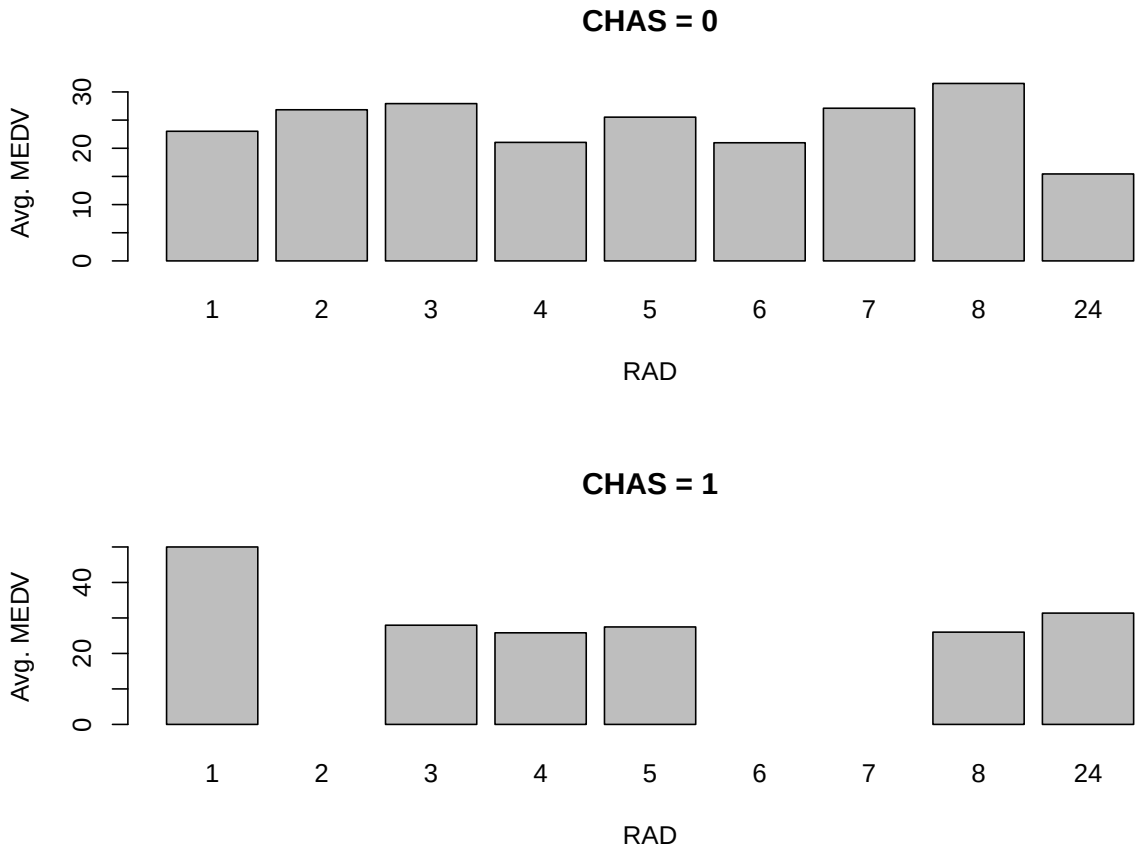

84

```
# alternative plot with ggplot
ggplot(data.for.plot) +
  geom_bar(aes(x = as.factor(RAD), y = `meanMEDV`), stat = "identity") +
  xlab("RAD") + facet_grid(CHAS ~ .)
```

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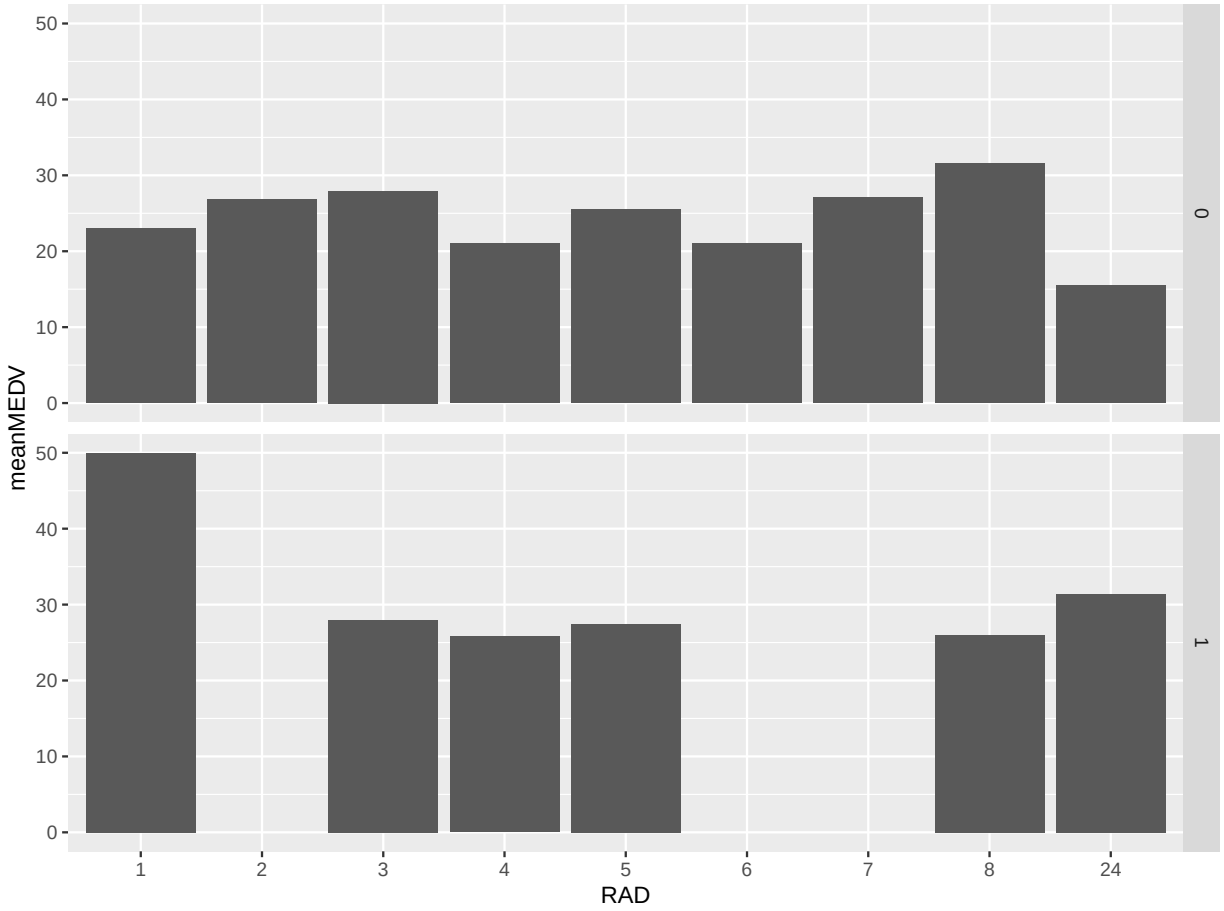

Figure 3.7

simple plot

```
# use plot() to generate a matrix of 4X4 panels with variable name on the diagonal,
# and scatter plots in the remaining panels.
plot(housing.df[, c(1, 3, 12, 13)])
```

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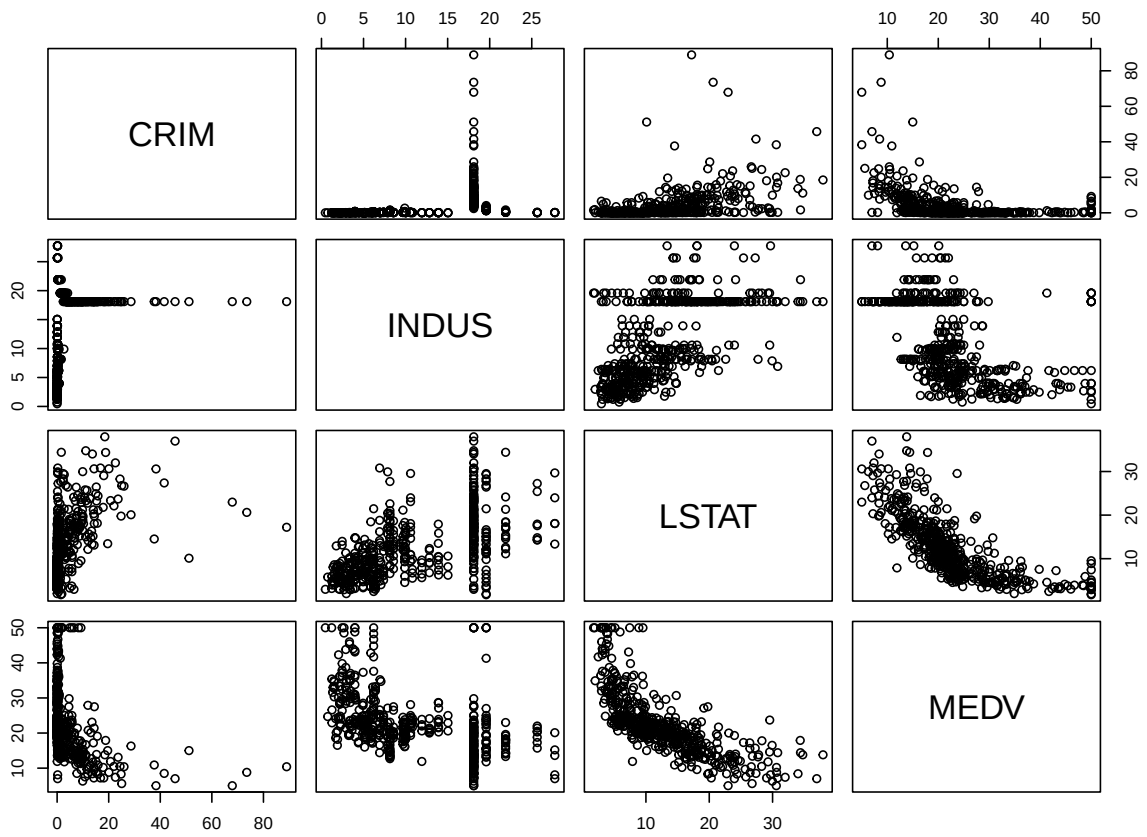

alternative, nicer plot (displayed)

```
library(GGally)
```

```
Registered S3 method overwritten by 'GGally':
  method from
  +.gg   ggplot2
```

```
ggpairs(housing.df[, c(1, 3, 12, 13)])
```

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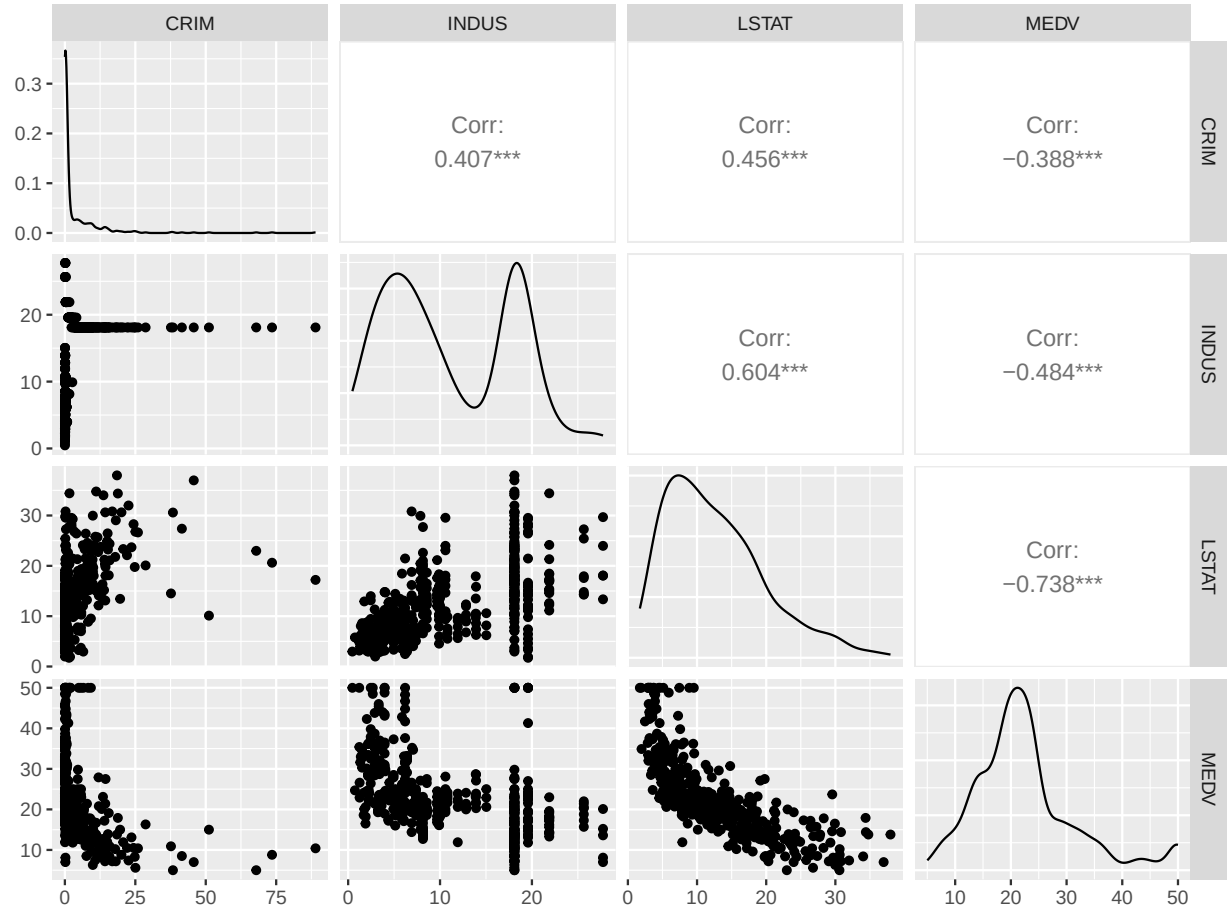

Figure 3.8

Simple plot MEDV CRIM

```
options(scipen=999) # avoid scientific notation

## scatter plot: regular and log scale
plot(housing.df$MEDV ~ housing.df$CRIM, xlab = "CRIM", ylab = "MEDV")
```

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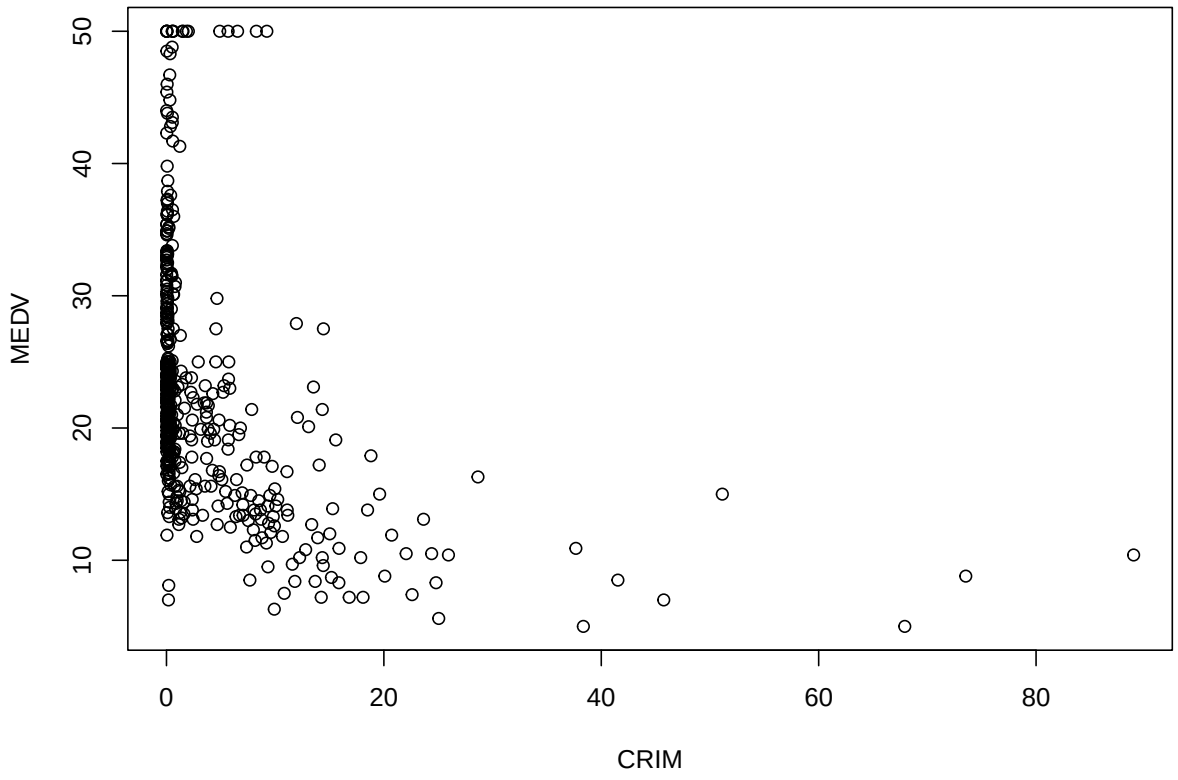

96

97 **to use logarithmic scale set argument log = to either 'x', 'y', or 'xy'.**

```
plot(housing.df$MEDV ~ housing.df$CRIM,
     xlab = "CRIM", ylab = "MEDV", log = 'xy')
```

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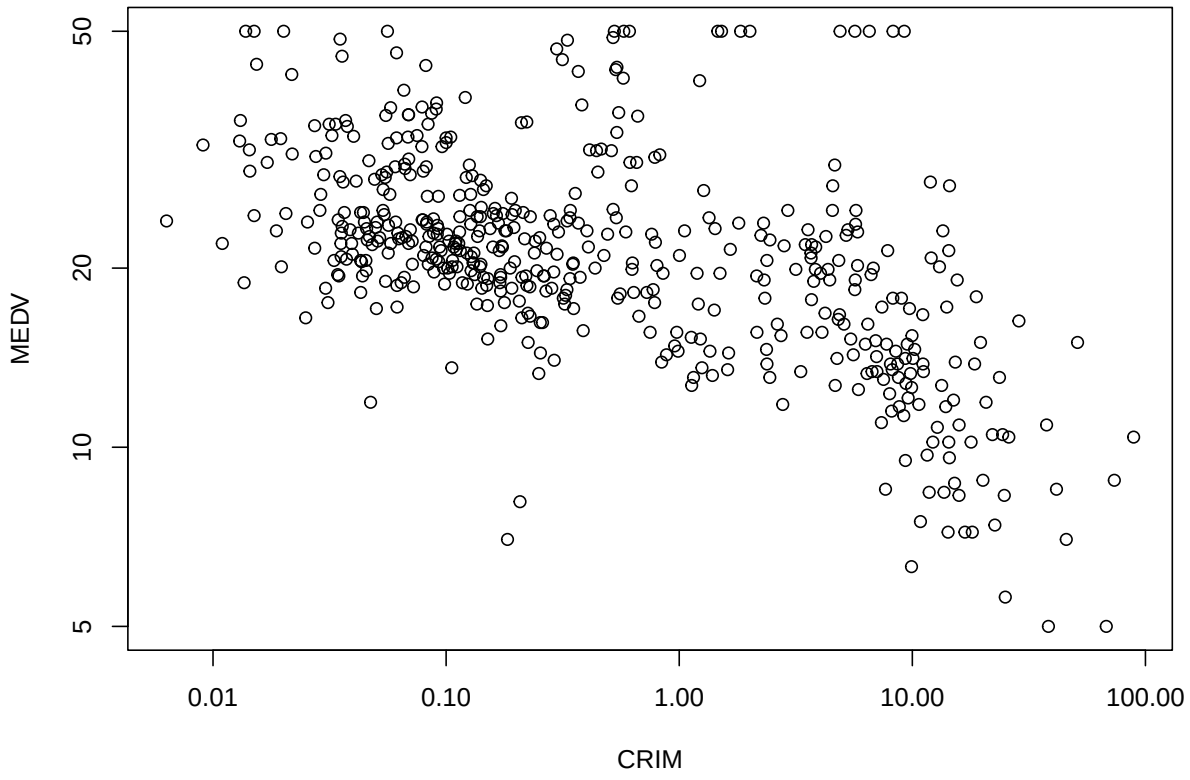

98

99 **alternative log-scale plot with ggplot**

```
library(ggplot2)
ggplot(housing.df) + geom_point(aes(x = CRIM, y = MEDV)) +
  scale_x_log10(breaks = 10^(-2:2),
                labels = format(10^(-2:2), scientific = FALSE, drop0trailing = TRUE)) +
  scale_y_log10(breaks = c(5, 10, 20, 40))
```

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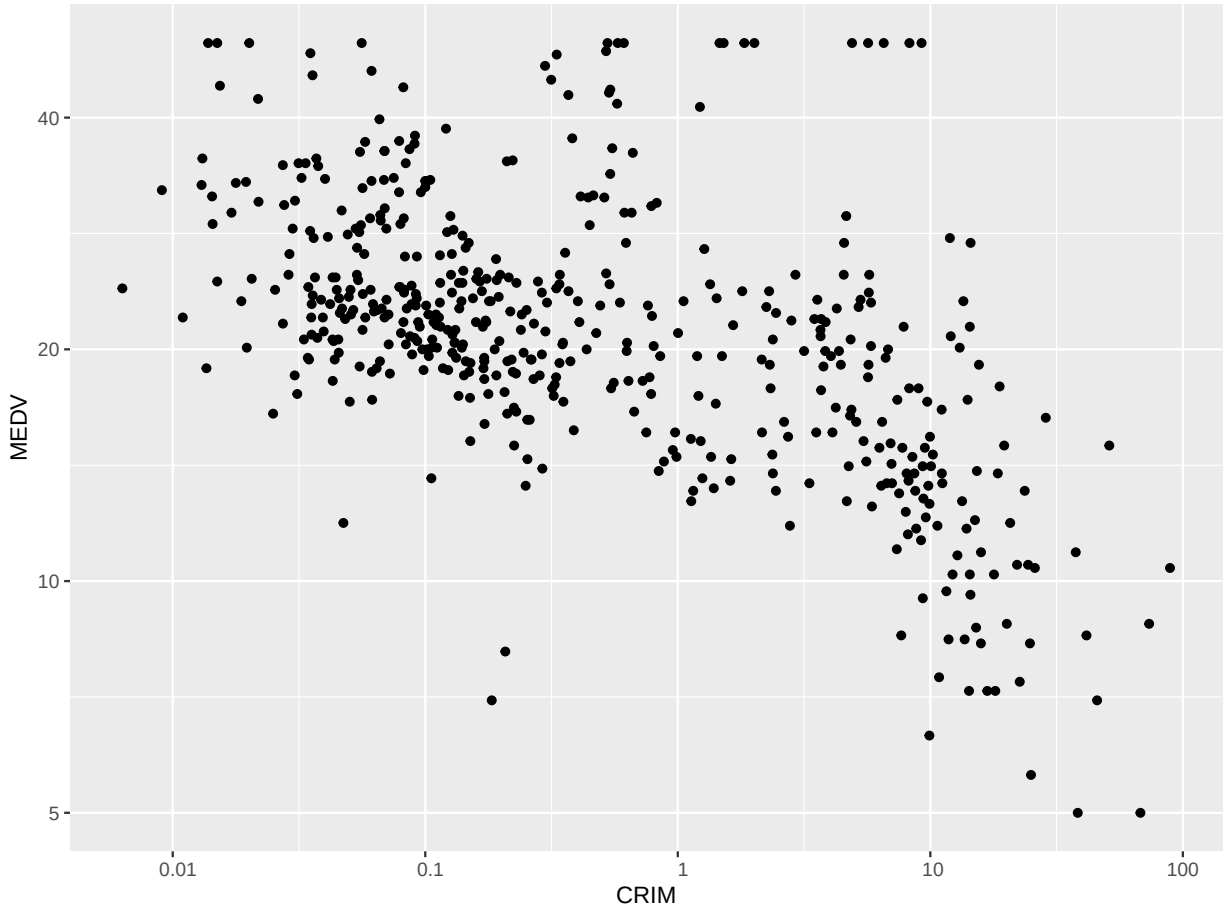

100

101 **boxplot: regular and log scale**

```
boxplot(housing.df$CRIM ~ housing.df$CAT..MEDV,
        xlab = "CAT.MEDV", ylab = "CRIM")
```

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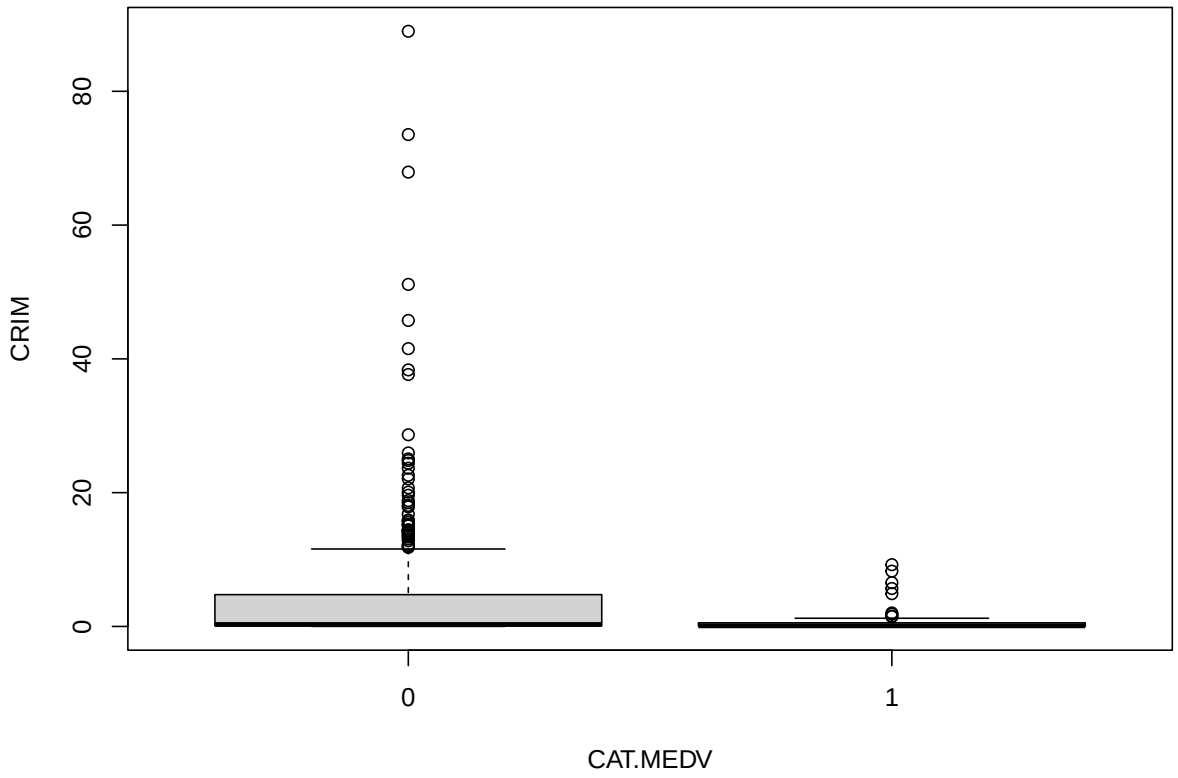

102

```
boxplot(housing.df$CRIM ~ housing.df$CAT..MEDV,
        xlab = "CAT.MEDV", ylab = "CRIM", log = 'y')
```

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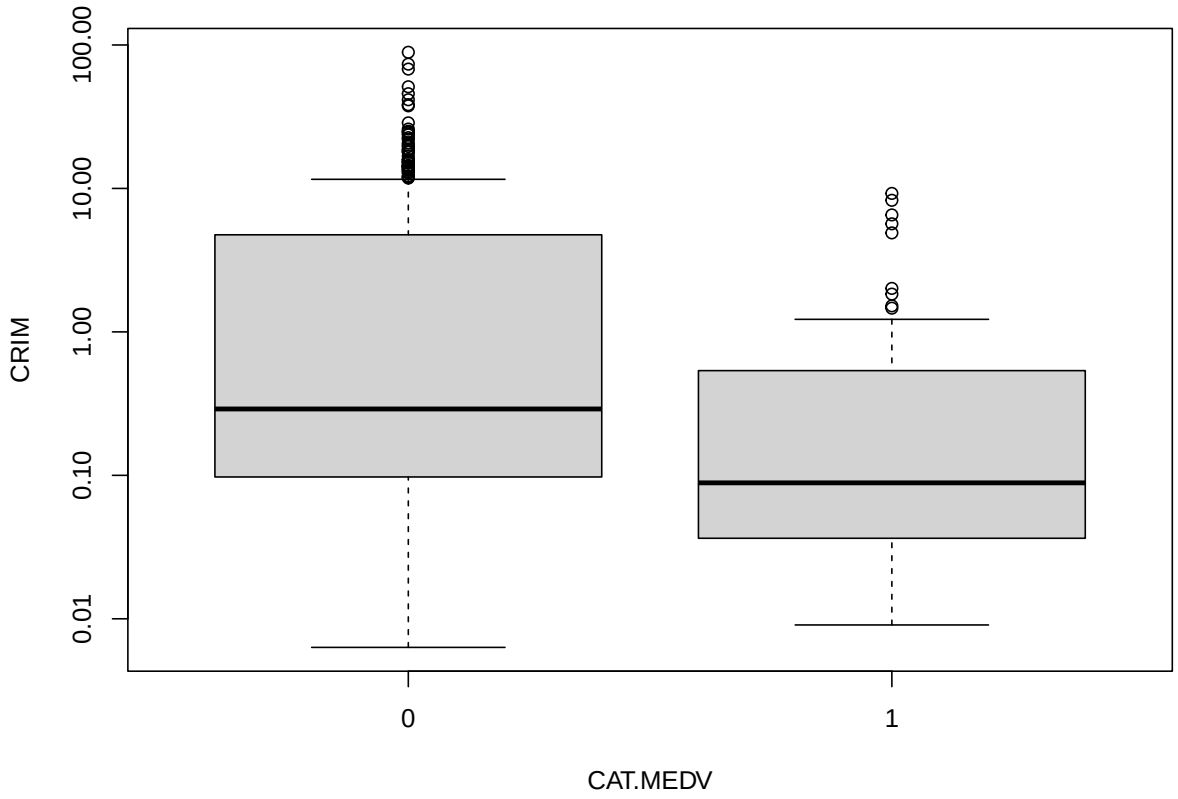

103

104 **Figure 3.9**

```
library(forecast)
Amtrak.df <- read.csv("data/Amtrak data.csv")
ridership.ts <- ts(Amtrak.df$Ridership, start = c(1991, 1), end = c(2004, 3), freq = 12)
```

105 **fit curve**

```
ridership.lm <- tslm(ridership.ts ~ trend + I(trend^2))
plot(ridership.ts, xlab = "Year", ylab = "Ridership (in 000s)", ylim = c(1300, 2300))
lines(ridership.lm$fitted, lwd = 2)
```

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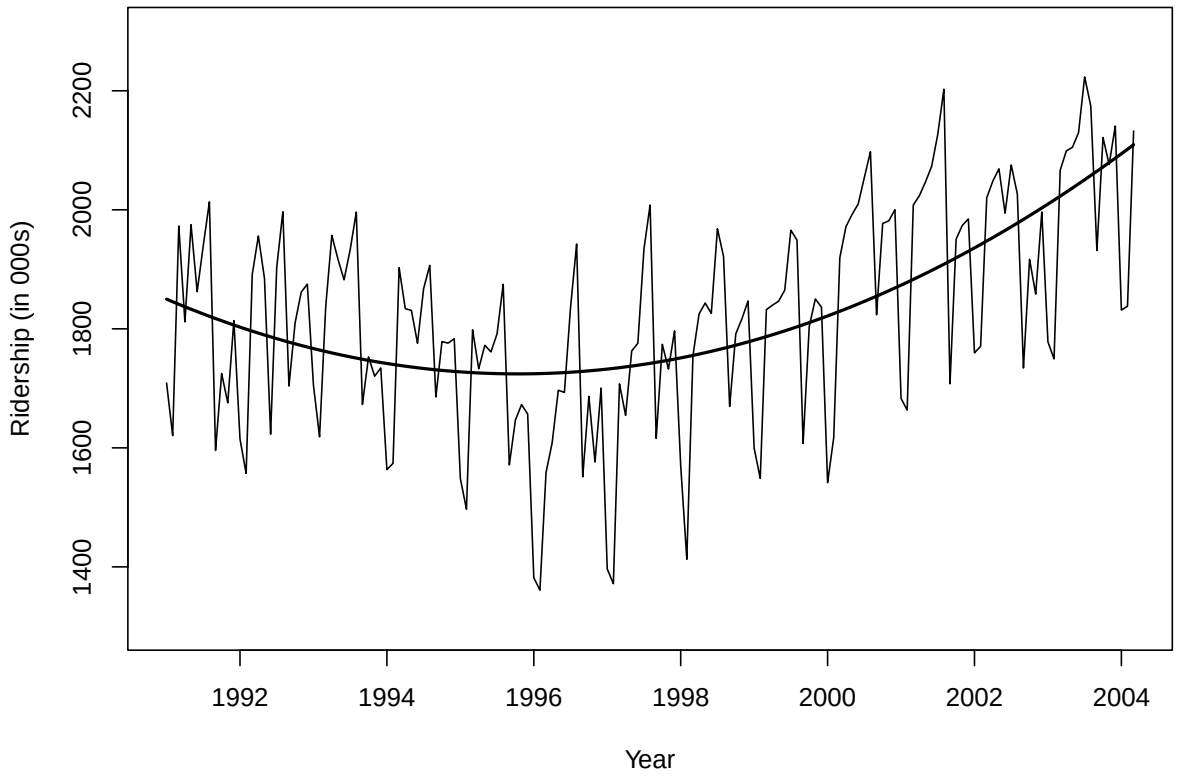

106

107 **alternative plot with ggplot**

```
library(ggplot2)
ggplot(Amtrak.df, aes(y = Ridership, x = Month, group = 12)) +
  geom_line() + geom_smooth(formula = y ~ poly(x, 2), method= "lm",
                            colour = "navy", se = FALSE, na.rm = TRUE)
```

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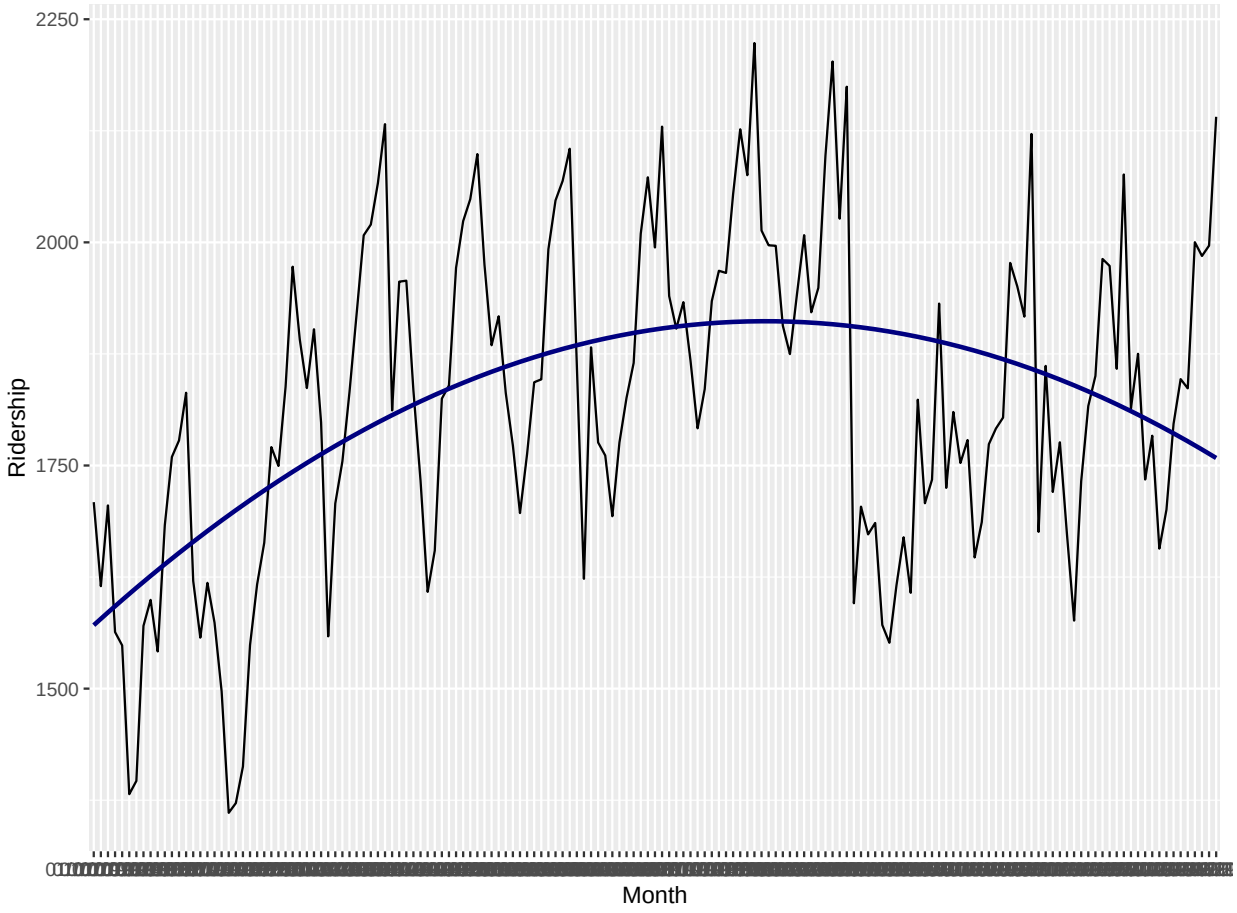

108

109 **Year**

```
ridership.2yrs <- window(ridership.ts, start = c(1991,1), end = c(1992,12))
plot(ridership.2yrs, xlab = "Year", ylab = "Ridership (in 000s)", ylim = c(1300, 2300))
```

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110

111 **Month**

```
monthly.ridership.ts <- tapply(ridership.ts, cycle(ridership.ts), mean)
plot(monthly.ridership.ts, xlab = "Month", ylab = "Average Ridership",
     ylim = c(1300, 2300), type = "l", xaxt = 'n')
## set x labels
axis(1, at = c(1:12), labels = c("Jan","Feb","Mar", "Apr","May","Jun",
                                 "Jul","Aug","Sep",  "Oct","Nov","Dec"))
```

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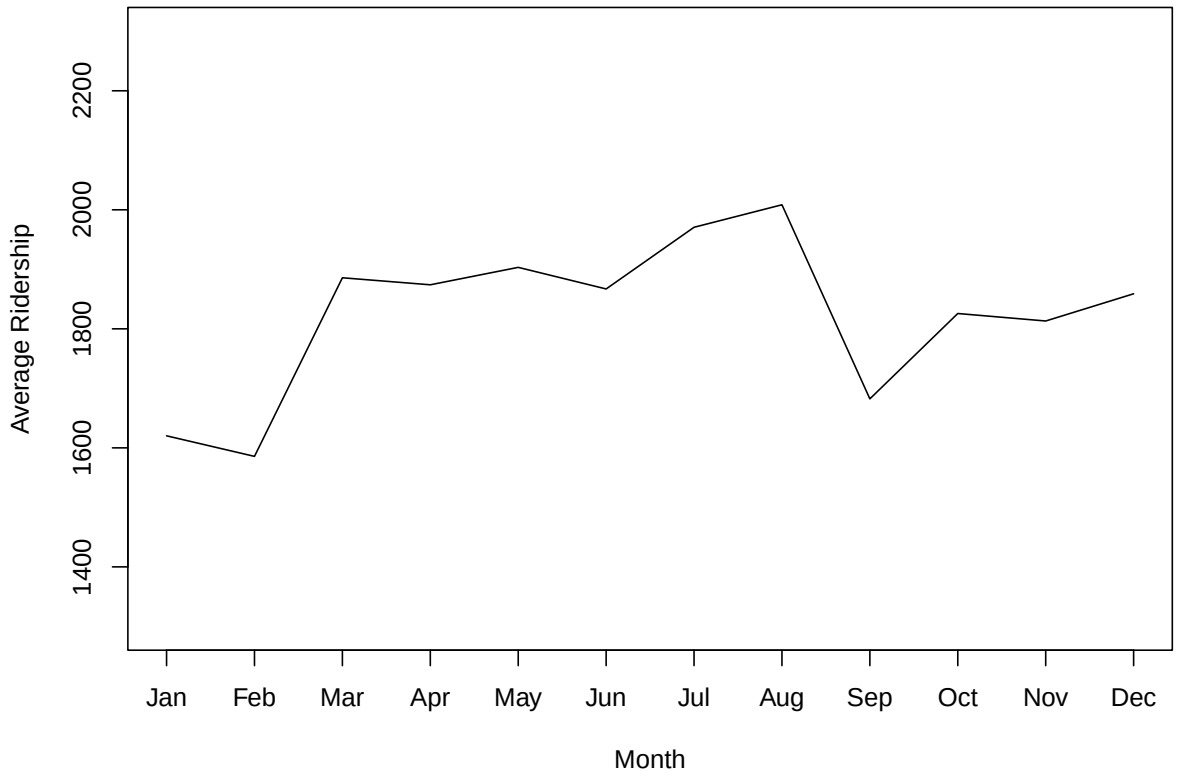

112

113 **Year(aggregated)**

```
annual.ridership.ts <- aggregate(ridership.ts, FUN = mean)
plot(annual.ridership.ts, xlab = "Year", ylab = "Average Ridership",
     ylim = c(1300, 2300))
```

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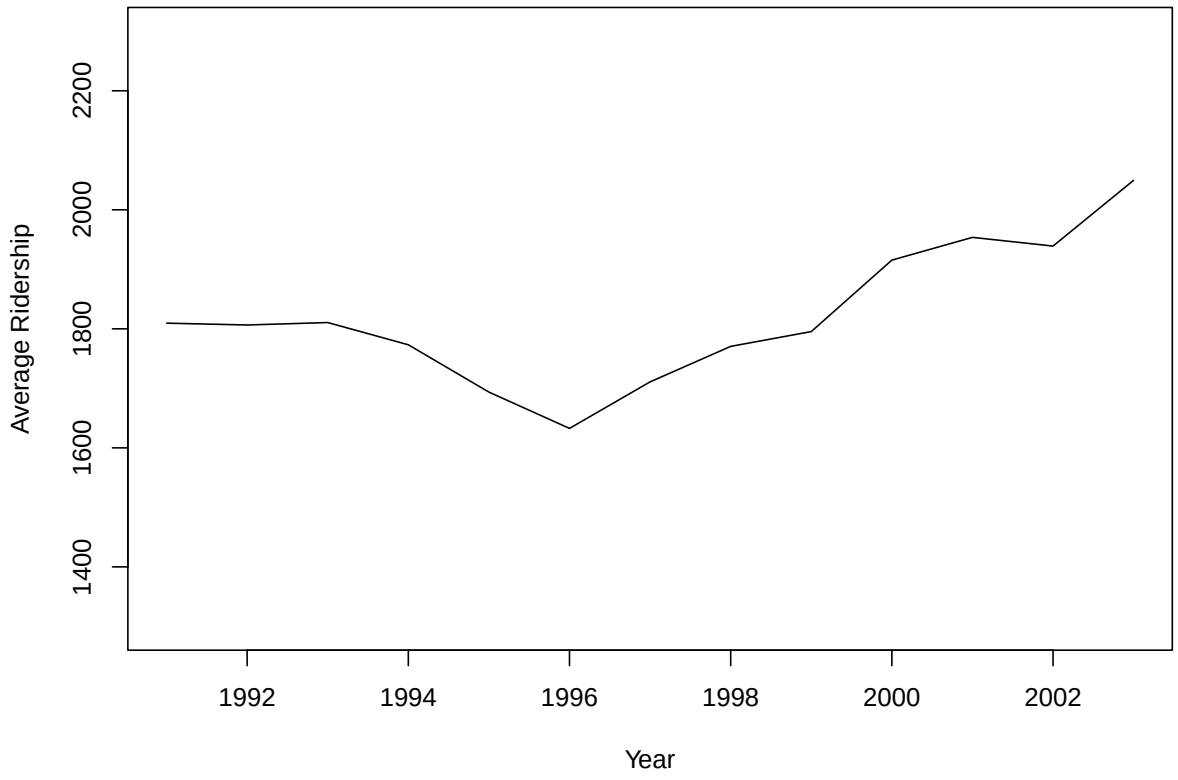

114

115 **Figure 3.10**

116 **Utilities sales**

```
utilities.df <- read.csv("data/Utilities.csv")

plot(utilities.df$Fuel_Cost ~ utilities.df$Sales,
     xlab = "Sales", ylab = "Fuel Cost", xlim = c(2000, 20000))
text(x = utilities.df$Sales, y = utilities.df$Fuel_Cost,
     labels = utilities.df$Company, pos = 4, cex = 0.8, srt = 20, offset = 0.2)
```

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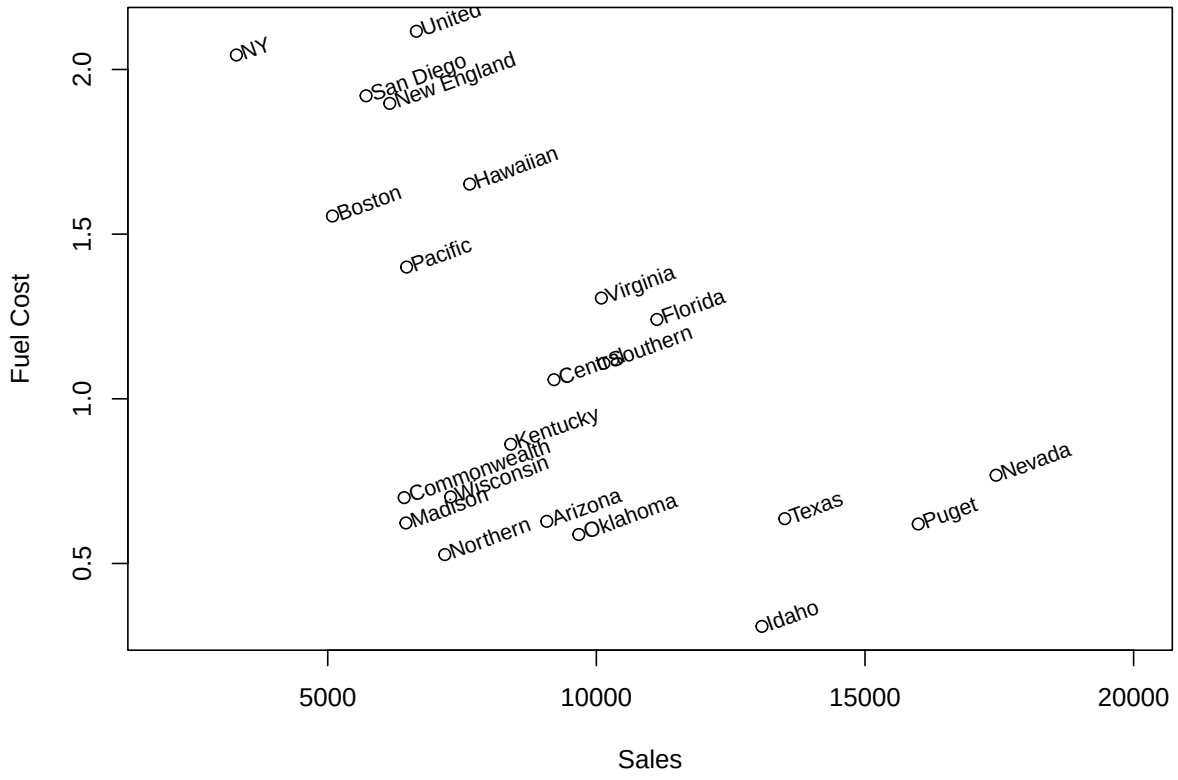

117

118 **alternative with ggplot**

```
library(ggplot2)
ggplot(utilities.df, aes(y = Fuel_Cost, x = Sales)) + geom_point() +
  geom_text(aes(label = paste(" ", Company)), size = 4, hjust = 0.0, angle = 15) +
  ylim(0.25, 2.25) + xlim(3000, 18000)
```

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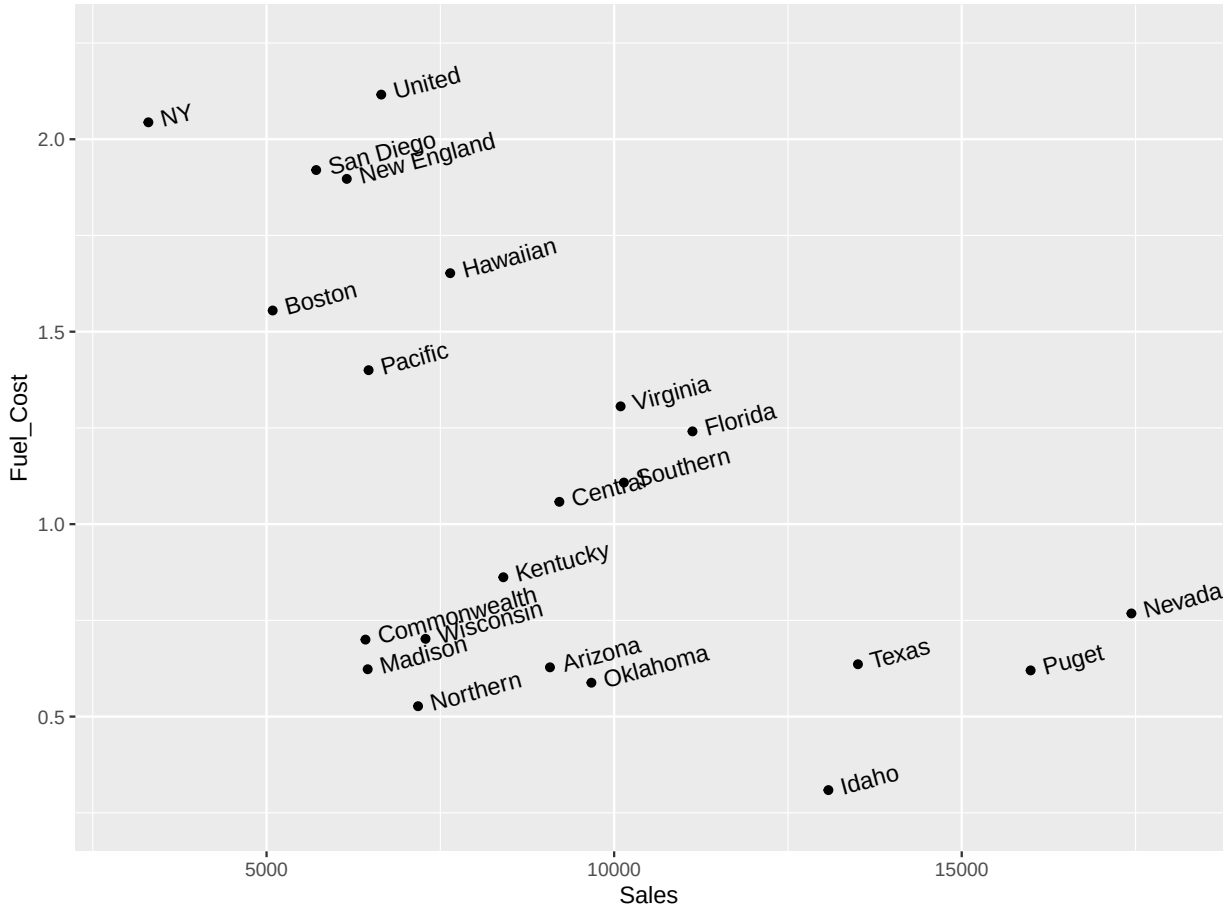

Figure 3.11

use function alpha() in library scales to add transparent colors

```
# use function alpha() in library scales to add transparent colors
universal.df <- read.csv("data/UniversalBank.csv")

library(scales)
```

```
Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard
```

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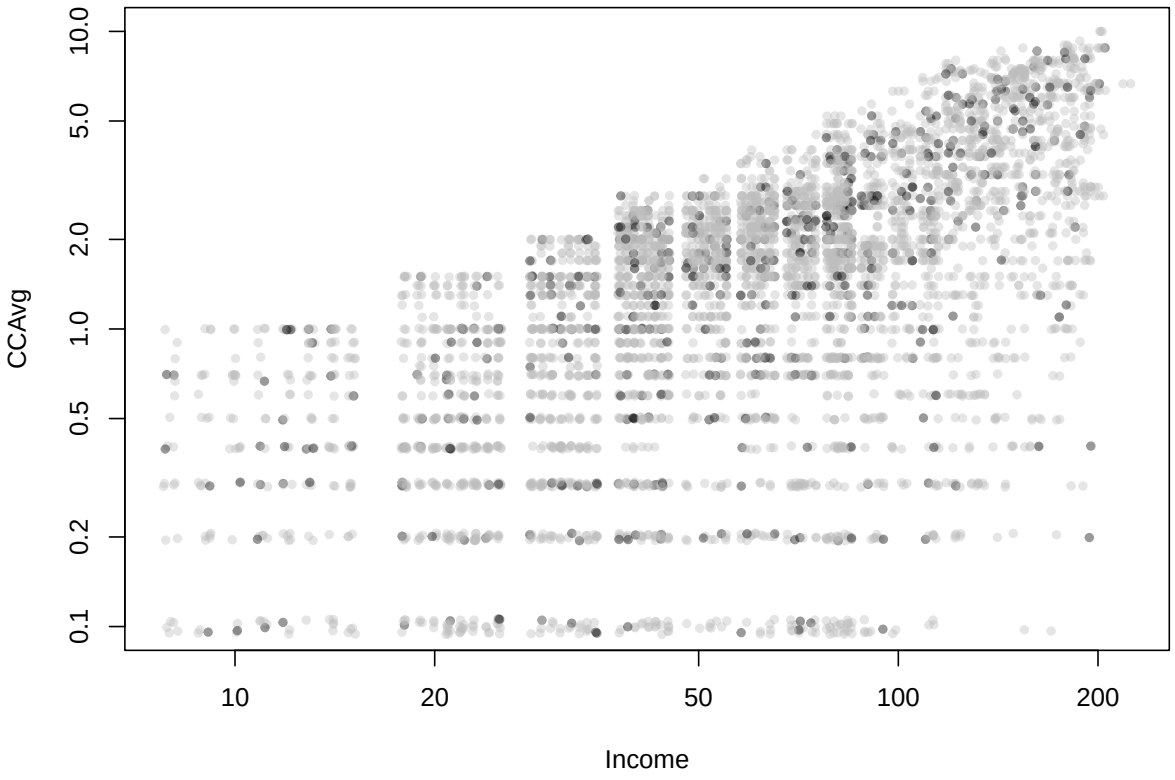
```
The following object is masked from 'package:readr':

    col_factor
```

```
plot(jitter(universal.df$CCAvg, 1) ~ jitter(universal.df$Income, 1),
     col = alpha(ifelse(universal.df$Securities.Account == 0, "gray", "black"), 0.4),
     pch = 20, log = 'xy', ylim = c(0.1, 10),
     xlab = "Income", ylab = "CCAvg")
```



alternative with ggplot

```
library(ggplot2)
ggplot(universal.df) +
  geom_jitter(aes(x = Income, y = CCAvg, colour = Securities.Account)) +
  scale_x_log10(breaks = c(10, 20, 40, 60, 100, 200)) +
  scale_y_log10(breaks = c(0.1, 0.2, 0.4, 0.6, 1.0, 2.0, 4.0, 6.0))
```

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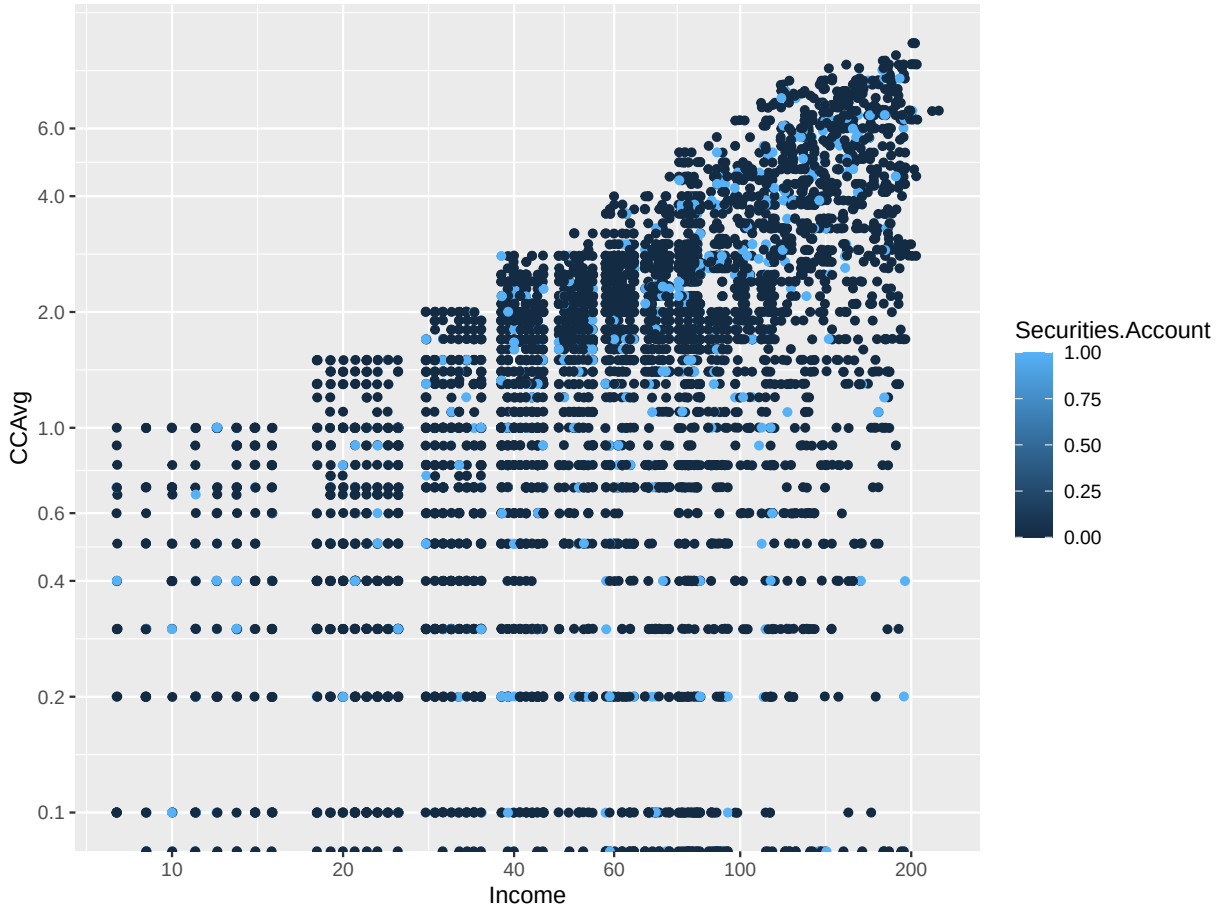

132

133 **Figure 3.12**

134 **Paracoord plot**

```
library(MASS)
```

135

136 ```
Attaching package: 'MASS'
```

137 ```
The following object is masked from 'package:dplyr':
```

138

139 ```
    select
```

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```
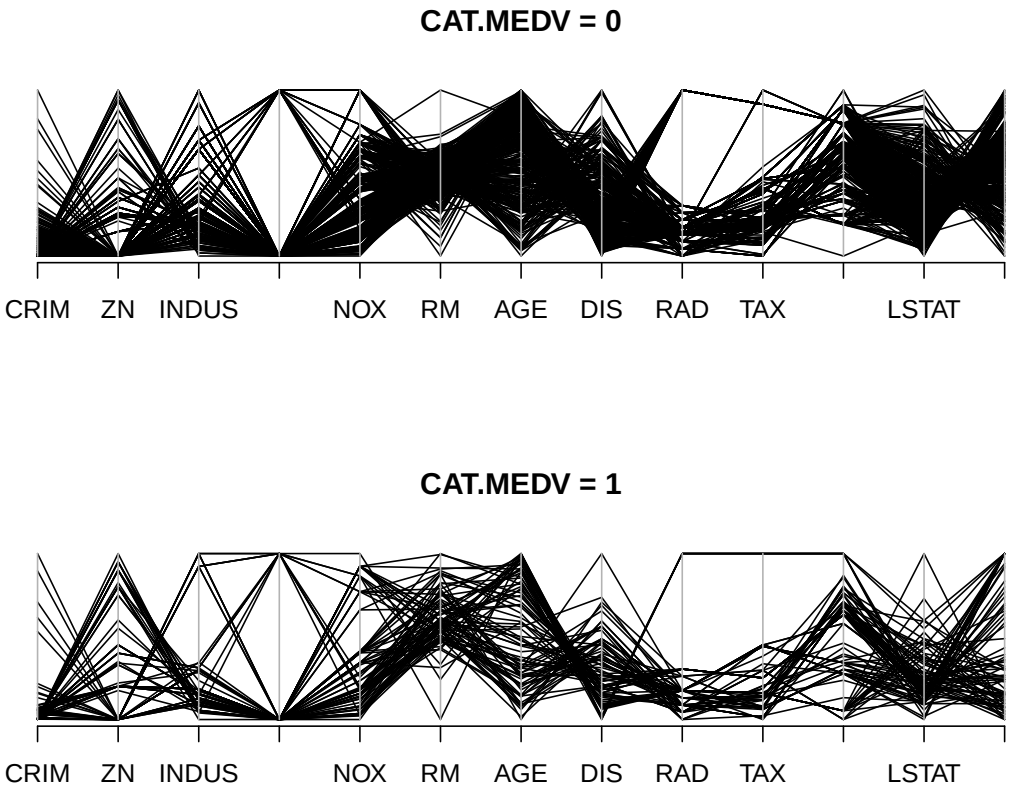

```
par(mfcol = c(2,1))
parcoord(housing.df[housing.df$CAT..MEDV == 0, -14], main = "CAT.MEDV = 0")
parcoord(housing.df[housing.df$CAT..MEDV == 1, -14], main = "CAT.MEDV = 1")
```



140

141 **Figure 3.14**

142 **Network**

```
library(igraph)
```

143

```
144 Attaching package: 'igraph'
```

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```
The following objects are masked from 'package:lubridate':

    %--%, union

The following objects are masked from 'package:dplyr':

    as_data_frame, groups, union

The following objects are masked from 'package:purrr':

    compose, simplify

The following object is masked from 'package:tidyr':

    crossing

The following object is masked from 'package:tibble':

    as_data_frame

The following objects are masked from 'package:stats':

    decompose, spectrum

The following object is masked from 'package:base':

    union
```

```r
ebay.df <- read.csv("data/eBayNetwork.csv")

# transform node ids to factors
ebay.df[,1] <- as.factor(ebay.df[,1])
ebay.df[,2] <- as.factor(ebay.df[,2])

graph.edges <- as.matrix(ebay.df[,1:2])
g <- graph.edgelist(graph.edges, directed = FALSE)
isBuyer <- V(g)$name %in% graph.edges[,2]

plot(g, vertex.label = NA, vertex.color = ifelse(isBuyer, "gray", "black"),
     vertex.size = ifelse(isBuyer, 7, 10))
```

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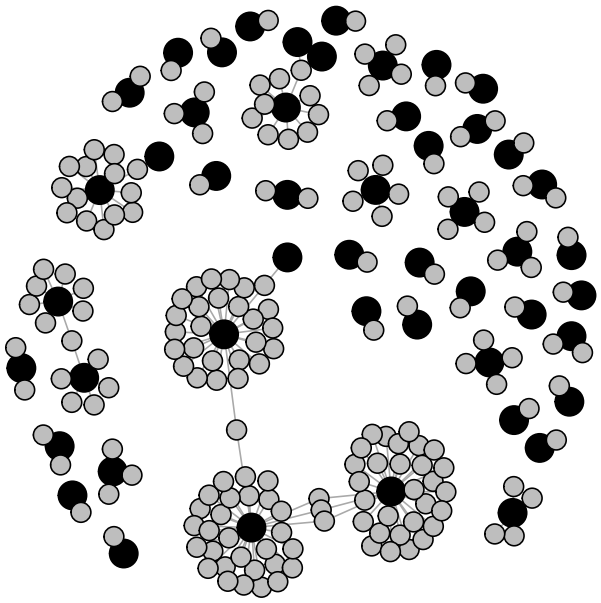

166

167 Figure 3.15

168 Treemap

```
library(treemap)
tree.df <- read.csv("data/EbayTreemap.csv")

# add column for negative feedback
tree.df$negative.feedback <- 1* (tree.df$Seller.Feedback < 0)

# draw treemap
treemap(tree.df, index = c("Category","Sub.Category", "Brand"),
        vSize = "High.Bid", vColor = "negative.feedback", fun.aggregate = "mean",
        align.labels = list(c("left", "top"), c("right", "bottom"), c("center", "center")),
        palette = rev(gray.colors(3)), type = "manual", title = "")
```

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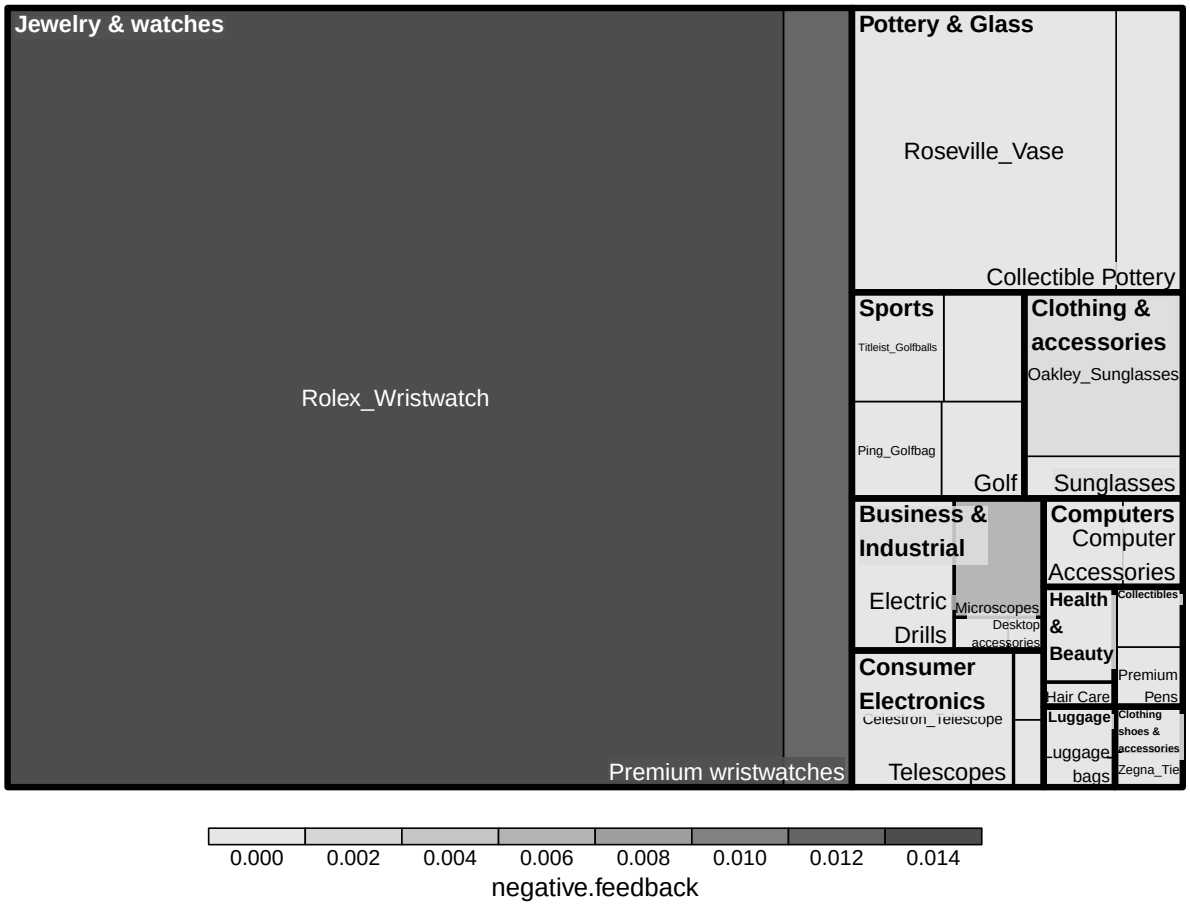

169

170 Figure 3.16

171 Map1

```
#library(ggmap)
#SCstudents <- read.csv("data/SC-US-students-GPS-data-2016.csv")
#Map <- get_map("Denver, CO", zoom = 3)
#ggmap(Map) + geom_point(aes(x = longitude, y = latitude), data = SCstudents,
#                        alpha = 0.4, colour = "red", size = 0.5)
```

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172 ## Figure 3.17

173 ### Map2

```
#library(mosaic)
#
#gdp.df <- read.csv("data/gdp.csv", skip = 4, stringsAsFactors = FALSE)
#names(gdp.df)[5] <- "GDP2015"
#happiness.df <- read.csv("data/Veerhoven.csv")
#
# gdp map
#mWorldMap(gdp.df, key = "Country.Name", fill = "GDP2015") + coord_map()
#
# eell-being map
#mWorldMap(happiness.df, key = "Nation", fill = "Score") + coord_map() +
#  scale_fill_continuous(name = "Happiness")
```

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